



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

## **SECP2523: DATABASE**

**SEM I 2025/2026**

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### **DATABASE DESIGN DESCRIPTION: CONCEPTUAL DATABASE DESIGN**

#### **FOR Hasta Travel Vehicle Rental Management System**

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## 1.0 Introduction

Hasta Travel and Tour SDN BHD is a vehicle rental company that serves exclusively to staff and students at Universiti Teknologi Malaysia (UTM). Facilitating seamless transportation for employees and students is the company's main goal. This project focuses on digitizing the company's current activities in an attempt to fit into modern digital norms. The project intends to simplify the leasing process by integrating a new digitalized system with their business model. This would reduce the administrative workload while also facilitating user access to services.

The company's current reliance on a manual booking method gives serious operational challenges. Under the current procedure, customers must complete a "WhatsForm" on the website and send individual WhatsApp inquiries. Time consuming availability checks and a slow response rate are just two of the inefficiencies caused by this manual method. Additionally, the absence of a centralized system raises the probability of human errors, such as misplaced communications or contradictory reservations, which combines both operational effectiveness and the general customer experience..

To address these challenges, this project proposes the development of a responsive web based vehicle rental management system. which will offer a real time dashboard displaying vehicle availability and dynamic pricing that adjusts based on seasonal demand. Secure data management will be handled through a MySQL database, ensuring the safe storage of user, staff, and vehicle information. Additionally, the system will automate communication by sending booking confirmations and invoices directly to the user's registered email address, eliminating the need for manual confirmation messages.

Firstly, the user will have a 24/7 real time vehicle availability update and an automated booking confirmation details sent through their emails. Next, the manager (system administration) will have a digital record keeping rather than needing to manually update their information and the workload of each staff will be reduced significantly. While established commercial platforms such as Penang Book, Wahdah and Socar offer fully digitalized vehicle rental services, the proposed system is distinct in its specific focus on the UTM ecosystem.

## 1.1 Definitions, Acronyms and Abbreviation

### 1.1.1 Definitions

Table 1.1.1.1 Definitions

| Term         | Definition                                                                                                                                                                        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin        | An authorized system user with elevated privileges to manage day-to-day operations including customer verification, booking approval, payment verification, and fleet management. |
| Manager      | The highest-level system user with full access to all modules, including financial reports, user role management, and system configuration.                                       |
| Blacklist    | A status assigned to customers who have outstanding payments, penalties, or other violations, restricting them from making new bookings until issues are resolved.                |
| Grace Period | A 30-minute buffer time after the scheduled return time before late penalties are applied.                                                                                        |
| Surcharge    | Additional fees applied for pickups or returns at non-standard locations or during specific time periods.                                                                         |

### 1.1.2 Acronyms

Table 1.1.2.1 Acronyms

| Acronym | Meaning                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------|
| API     | A set of protocols and tools that allow different software applications to communicate with each other. |
| CC      | The engine displacement or volume of an engine's cylinders, measured in cubic centimeters.              |
| IC      | Malaysian national identity card (MyKad) required for customer verification.                            |
| JPJ     | The Road Transport Department of Malaysia, responsible for vehicle registration and licensing.          |
| UTM     | Universiti Teknologi Malaysia                                                                           |

### 1.1.3 Abbreviation

Table 1.1.3.1 Abbreviation

| Abbreviation | Meaning                              |
|--------------|--------------------------------------|
| IC No        | Identity Card Number (00000-00-0000) |
| Matric No    | Matric Number (X00XX0000)            |

## 2.0 DFD (to-be)

### 2.1 Context Diagram

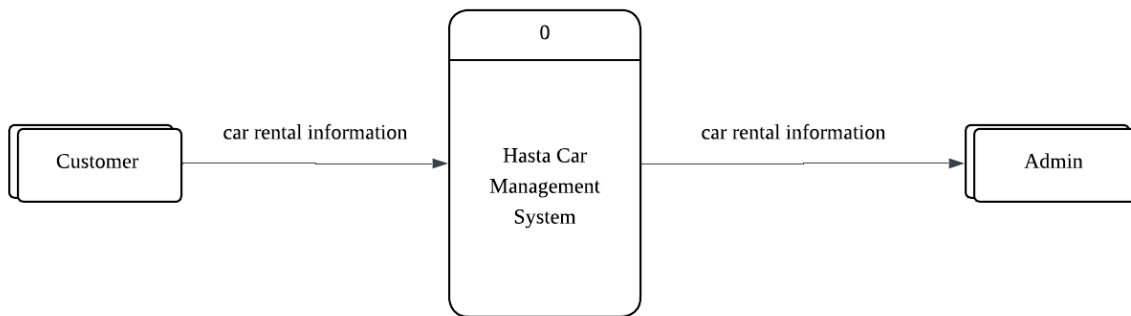


Figure 2.1: Context Diagram

### 2.2 Diagram 0

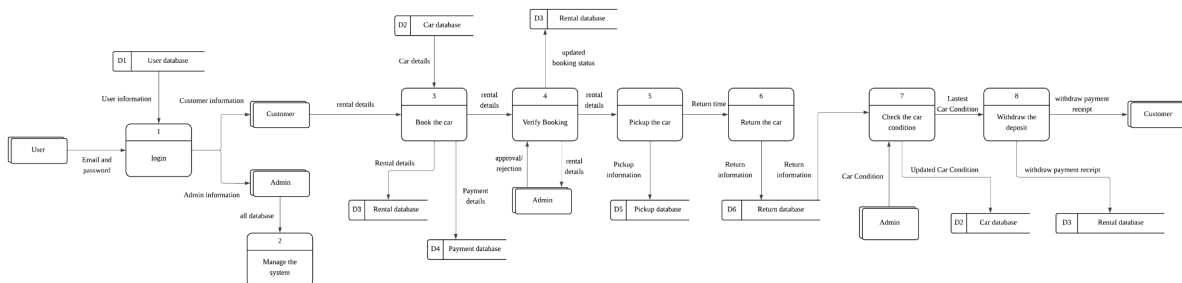


Figure 2.2: Diagram 0

### 2.3 Child Diagram

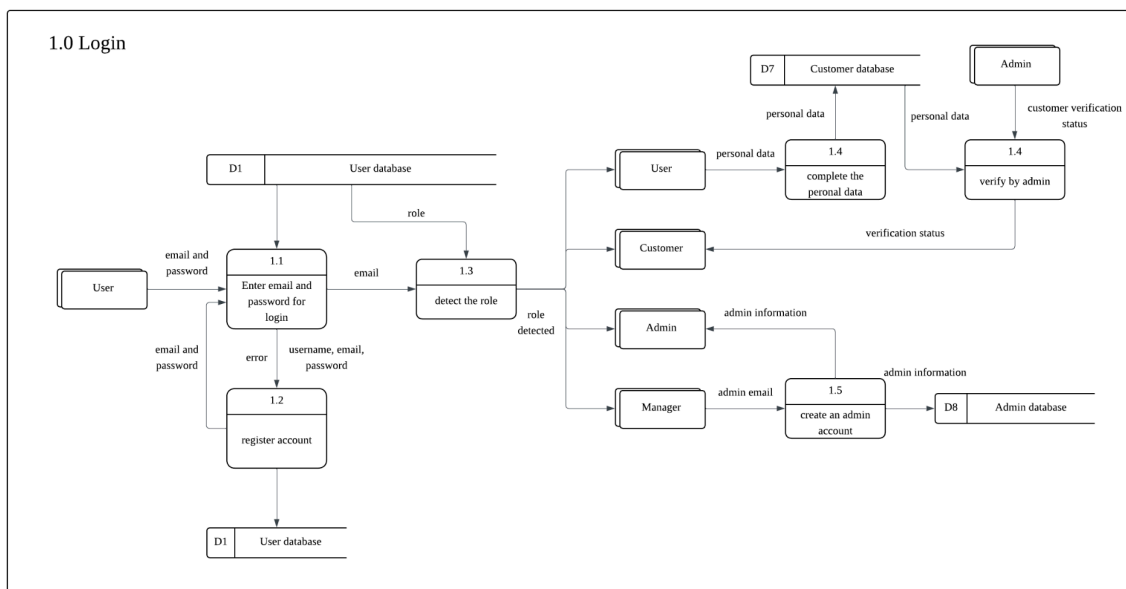


Figure 2.3.1 Child Diagram for Process 1: Login

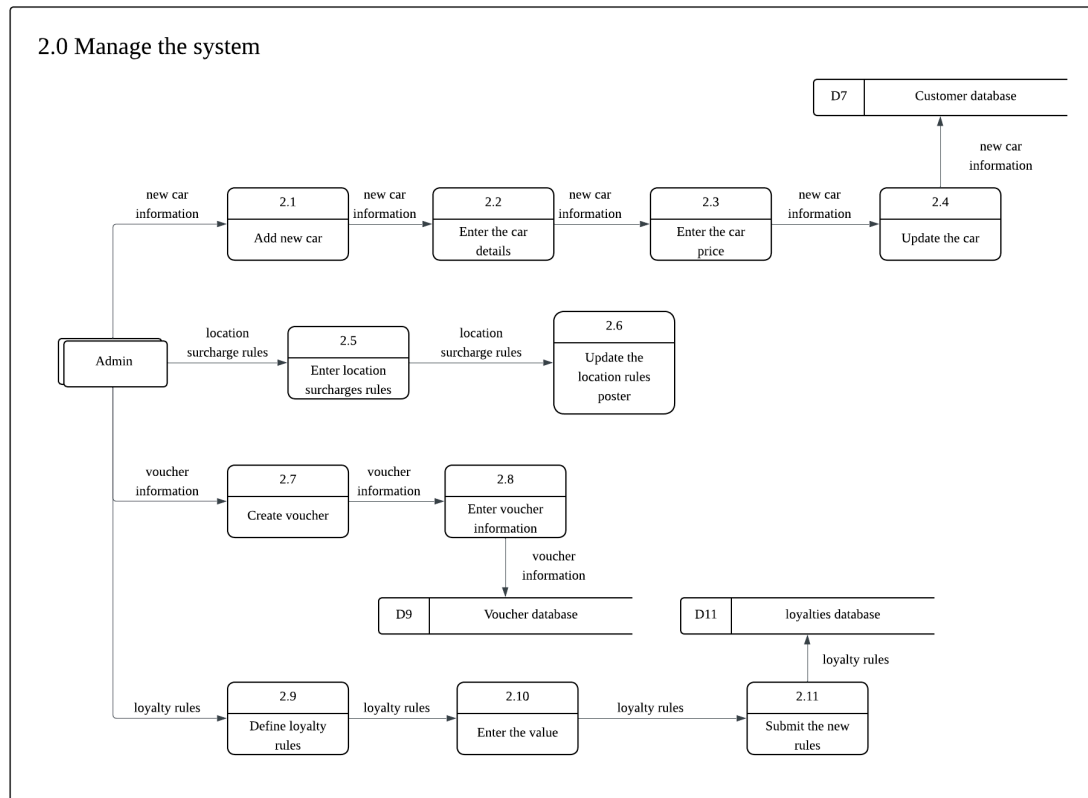


Figure 2.3.2 Child Diagram for Process 2: Manage the System

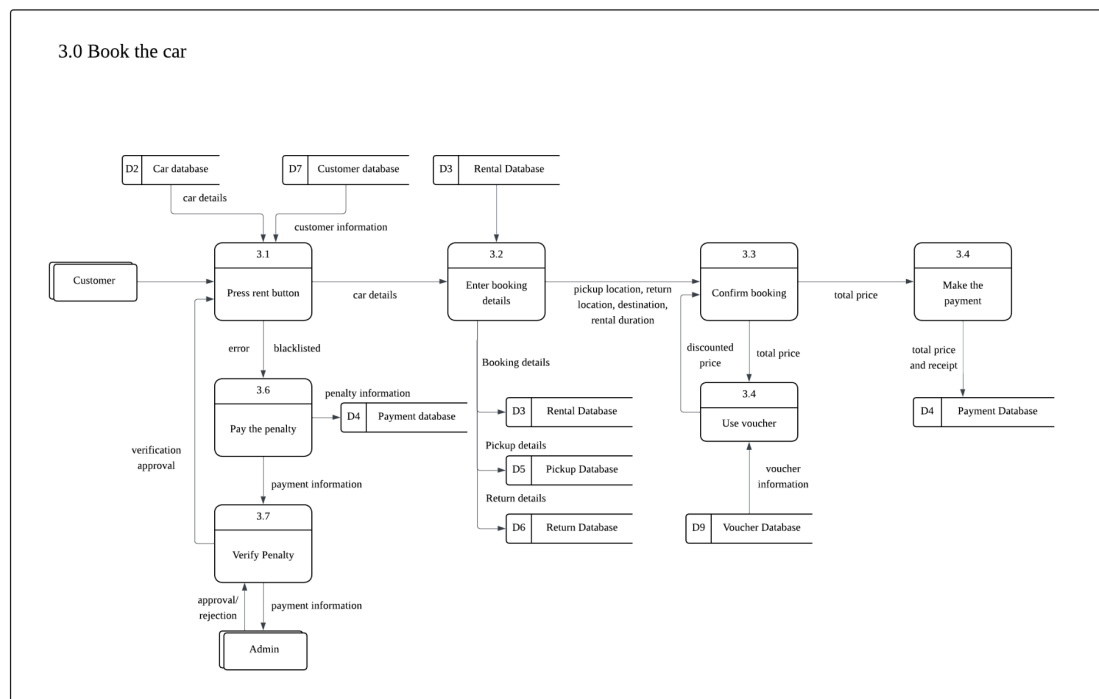


Figure 2.3.3 Child Diagram for Process 3: Book the Car

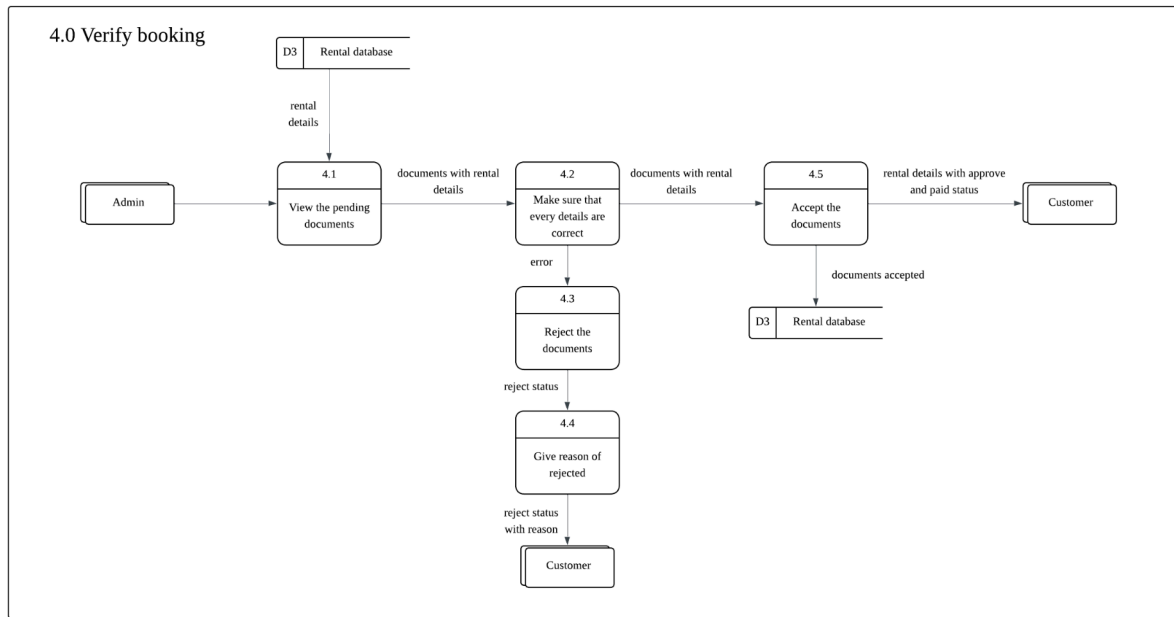


Figure 2.3.4 Child Diagram for Process 4: Verify Booking

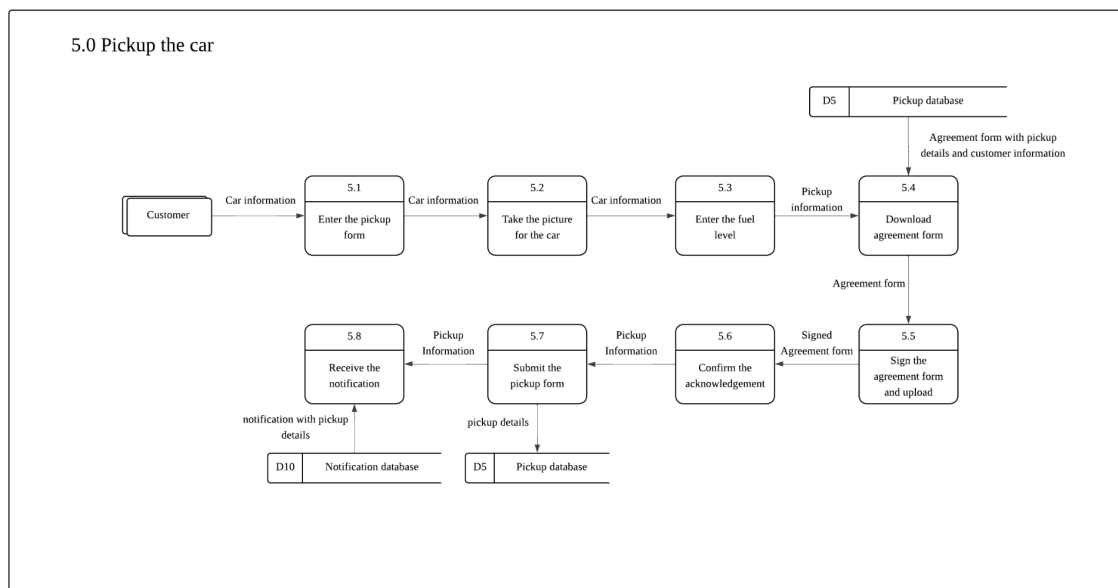


Figure 2.3.5 Child Diagram for Process 5: Pickup the Car

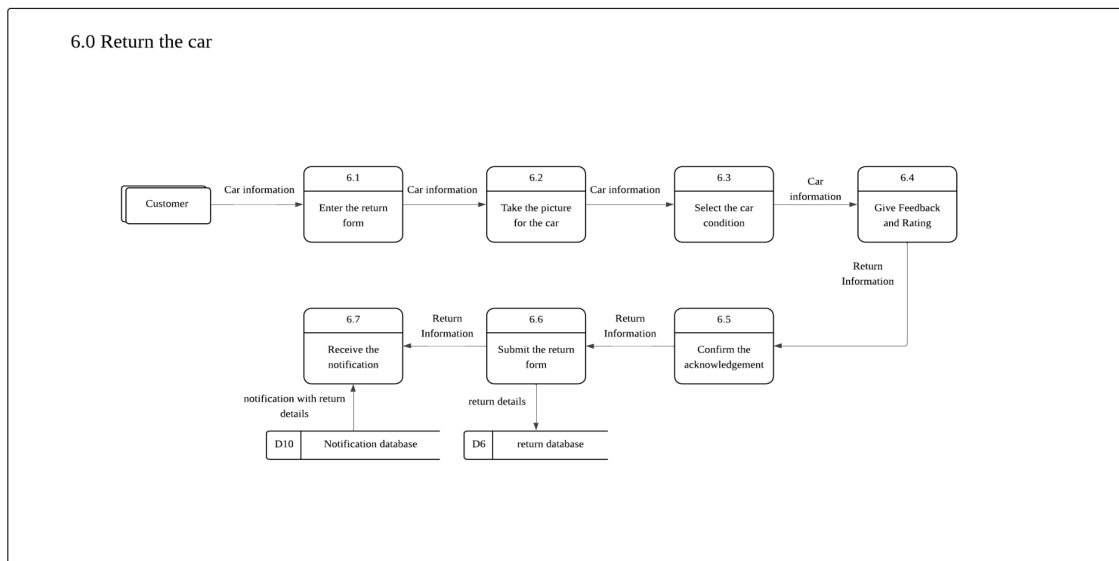


Figure 2.3.6 Child Diagram for Process 6: Return the Car

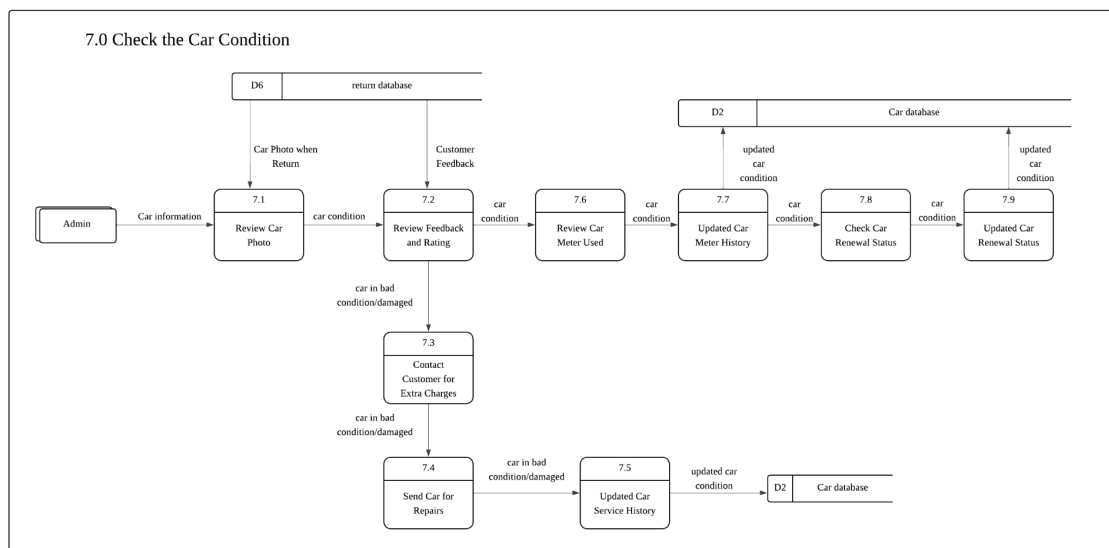


Figure 2.3.7 Child Diagram for Process 7: Check the Car Condition

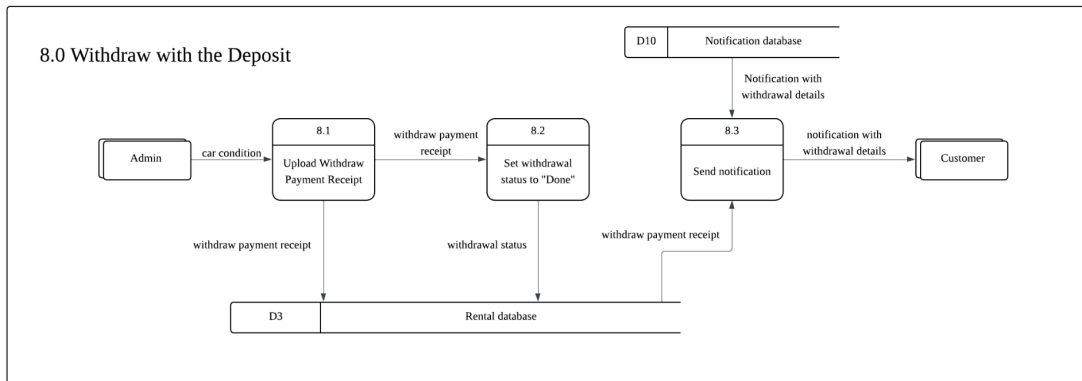


Figure 2.3.8 Child Diagram for Process 8: Withdraw the Deposit

### **3.0 Data and Transactional Requirements**

#### **3.1 Proposed business rule**

This section outlines both the business rules and also the data and transactional requirements that the new Hasta Travel and Tours System need to attain.

##### **3.1.1 User Management Module**

1. All new public users are registered with the customer role by default
2. Customers must complete their specific profile data which are driving license, Identification Card (IC) or passport and matric card before they are permitted to book a vehicle
3. Uploaded documents require manual approval by an admin or manager
4. Admins or managers can approve or reject these documents and a reason is needed when rejection occurs
5. When the customers change the IC number or matric no, they have to re-upload the related documents
6. Booking privileges are blocked until documents are approved
7. Customers with outstanding penalties are marked as blacklisted and cannot make new bookings until they upload a payment receipt which is then verified by an admin

##### **3.1.2 Fleet Management Module**

1. The system designates vehicles in two primary states which are 'active' for those currently in the operational fleet and 'retired' for those permanently inactive or the owner does not want to rent the vehicle anymore
2. Vehicle availability is tracked dynamically which 'available' indicates readiness for booking while 'unavailable' indicates the vehicle is currently rented or disabled
3. A maintenance status overrides standard availability which prevents bookings when a vehicle is undergoing repairs
4. Rental prices are defined individually for each vehicle across standard durations which are one hour, three hours, five hours, twelve hours and one day. The system also applies different pricing tiers for weekends versus weekdays to accommodate demand fluctuations
5. The system records detailed owner information for each vehicle
6. Critical documents such as the vehicle grant, road tax and insurance cover note are uploaded and tracked digitally
7. The system actively tracks expiry dates for road tax, insurance and other recurring obligations. Admin can set due date and receive alerts for upcoming renewals
8. Service records log details of maintenance work, vendor name, cost and date while meter updates records tracks the odometer readings periodically to monitor vehicle usage

##### **3.1.3 Booking Management Module**

1. Customers must have a complete profile with all approved uploaded documents before they can initiate a booking

2. The system enforces a conflict check to strictly prevent any booking that might overlap with an existing pickup or return for the selected vehicle
3. The system applies additional fees for pickups or returns that occur at designated surcharge locations such as outside campus or during specific surcharge times. Surcharge rules are configurable via the admin and manager panel
4. Customers are only permitted to cancel a confirmed booking if the scheduled pickup time is more than 24 hours away.
5. Cancellations are eligible for a refund which is processed manually within 3 to 5 business days
6. If a booking using a voucher is cancelled, the voucher will be automatically restored for future use
7. Both pickup and return processes require a digital handover where the customer must upload four angle photos of the vehicle, a dashboard photo and note any damage
8. Fuel levels are recorded at both ends of the transaction to calculate penalties
9. Customers must digitally acknowledge the rental agreement at pickup
10. Customers are required to fill the ratings and feedback at return
11. Late return is calculated only after a 30-minute grace period from the scheduled return time and RM50 would be charged per hour
12. Fuel shortage penalties are applied based on the number of fuel bars missing as recorded during inspection which are RM10 per missing fuel bar
13. The penalties are subtracted from the deposit. If the penalty is less than deposit, the remaining deposit is refundable

#### **3.1.4 Payment Management Module**

1. A deposit is mandatory and collected for every single booking iteration
2. Admin must visually verify the uploaded receipt image before a booking can be confirmed
3. Rejected receipts will trigger a re-upload request or booking cancellation. Only “incorrect amount” triggers a refund workflow while other reasons typically require re-upload
4. The system records the specific admin ID who verified each payment for accountability

#### **3.1.5 Loyalty and Rewards Module**

1. Customers earn 1 loyalty stamp for every completed rental transaction with a duration of 9 hours or more
2. Customers can redeem collected stamps for vouchers based on defined tiers. Redemption immediately deducts the stamp count from the customer’s balance
3. Vouchers can optionally grant a percentage discount or specific free hours
4. A voucher can only be used once
5. If the booking is cancelled or rejected, the voucher will be automatically restored for future use
6. Admin and manager can assign voucher to the customers

### **3.1.6 Report and Analytics Module**

1. Admin dashboard focuses on operational or daily tasks while manager dashboard focuses on strategic performance
2. Revenue is aggregated into Daily (Revenue vs Expenses) and Monthly (Total Sales Count and Amount) views
3. The dashboard has fleet status overview which shows whether the fleet are available, rented or under maintenance
4. Feedback Scores view the vehicle rating especially those ratings below 3 stars to alert admin and manager

## **3.2 Proposed Data & Transactional Requirements**

### **3.2.1 User Management Module**

User Module serves as the centralized identity management component of the Hasta Travel & Tours Rental Management System. It securely manages all registered entities including Customers, Admin and Manager. Moreover, it controls authentication, authorization and profile information.

#### **3.2.1.1 Data Requirements**

##### **Users**

Hasta Travel Vehicle Rental Management System maintains user accounts for all individuals accessing the platform. Each user is uniquely identified by ID and must register with a valid and unique email address with password. Stored data includes, ID, email, username, password, role and account status.

The user role attribute determines the level of system access and dashboard features available after login. Roles include customer, admin and manager. User roles are not fixed at registration and may be updated by authorized personnel. A Manager has the authority to update a user's role directly within the User table, for instance, upgrading a Customer to a Manager. This design enables flexible role management without requiring the creation of new user accounts.

##### **Customers**

Customer records represent Universiti Teknologi Malaysia (UTM) students or staff using the booking service. Each customer is uniquely identified by a customer ID. Stored data includes customer ID, full name, nationality, matric or staff card image, matric or staff number, identity card or passport image, identity card or passport number, gender, faculty, residential college, address, phone number, driving license image, driving license expiry date, bank name, bank account, emergency contact information including name, relationship with customer and phone number, document verification status and customer status. Customer data is linked to the corresponding user account.

##### **Admin**

Admin members are system users with additional privileges and created through Manager actions. Each admin is uniquely identified by an admin ID. Stored data includes admin ID, phone number, position and admin status. Admin accounts are linked to user credentials for authentication.

### **3.2.1.2 Transaction Requirements**

#### **Data Entry**

1. Register a new user account with default role Customer
2. Promote an existing user to Manager by updating the role attribute
3. Create Admin account through Manager authorization

#### **Data Update/Deletion**

1. Update a user's role, for example, Customer to Manager
2. Activate or deactivate Admin privileges
3. Update user credentials and profile information
4. Update document verification status from pending to approved or rejected
5. Deactivate user accounts without deleting historical records

#### **Data Queries**

1. Retrieve user role during login to determine access rights
2. Load role-specific dashboard and permissions
3. Check email uniqueness during user registration
4. Handle password reset requests using registered email
5. Verify Manager authorization before allowing admin creation
6. List users based on role for manager management

### 3.2.1.3 Local Conceptual Data Model (ERD)

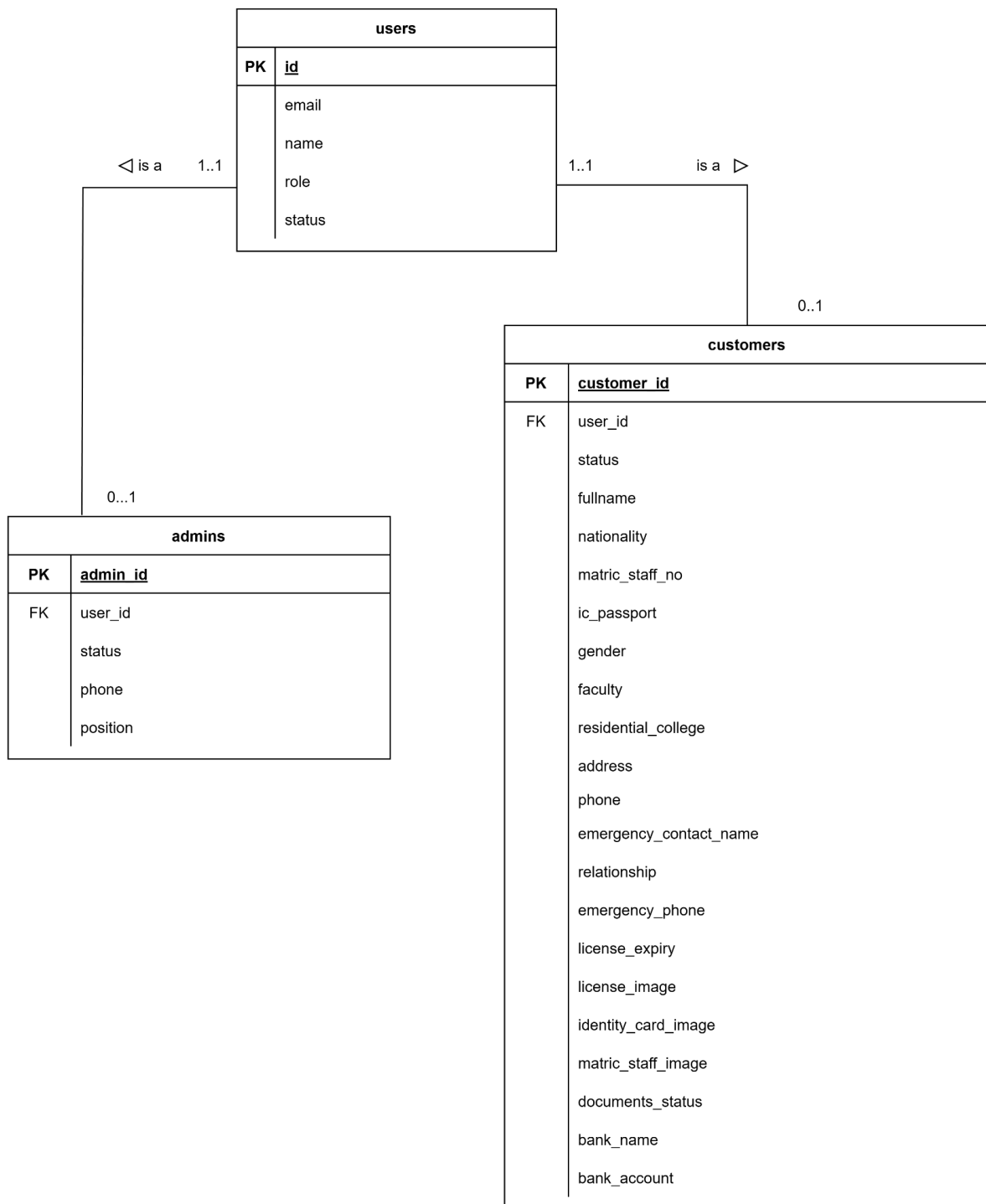


Figure 3.2.1.3.1 Local Conceptual Data Model for User Management Module

### **3.2.2 Fleet Management Module**

This section outlines the vehicle module transaction requirements, which manage vehicle information, availability and status in the Hasta Travel & Tours Vehicle Rental Management System.

#### **3.2.2.1 Data Requirements**

##### **Vehicle**

Each vehicle is uniquely identified by vehicle plate number. Stored data includes general information namely, plate number, model, colour, seat, vehicle mileage, production year, engine, engine CC, fuel type, transmission type, drive type, maximum speed, torque and power, pricing rules, manufacturer, operating state and owner details such as name, identity card number, age, gender, race, citizenship, grant image path, roadtax image path, cover note image path and their respective expiry dates.

Operational data includes status, availability status and maintenance status. Status defines whether the vehicle is a valid member of the fleet that can be rented or not while availability status and maintenance status are used to control booking and fleet management activities.

##### **Vehicle Photos**

The system stores multiple photos for each vehicle to provide visual reference for system users. Each photo record is uniquely identified by a photo ID and is linked to a specific vehicle plate number. Stored data includes photo ID, vehicle plate number and label. This separation allows multiple images to be associated with a single vehicle without duplicating vehicle data.

##### **Renewal Reminder**

The renewal reminder records are used to manage expiry-related vehicle documents such as road tax and insurance. Each renewal reminder is uniquely identified by a reminder ID and is linked to a specific vehicle. The data stored for each renewal reminder includes reminder ID, service type (such as insurance or road tax), due date, reminder notes and reminder status. This separation allows a vehicle to have multiple renewal reminders and enables the system to track upcoming expiries without affecting core vehicle data.

##### **Service History**

The system stores maintenance and meter update information for vehicles in the system. Each service history record is uniquely identified by a service ID and is linked to a specific vehicle. Stored data includes service ID, vehicle name, completed date, current odometer reading, previous odometer reading, previous odometer update date, service type, service vendor, service description and service amount. This supports multiple service and meter update records for the same vehicle, allowing the system to maintain a complete history of vehicle maintenance and mileage updates.

### **3.2.2.2 Transaction Requirements**

#### **Data Entry**

1. Register a new vehicle in the admin or manager site of the system with complete technical, legal and operational details
2. Upload one or more photos for a vehicle
3. Define hourly rental prices for a vehicle based on different time durations
4. Insert a renewal reminder record and record the service type, due date and optional notes
5. Insert a new service history record when a vehicle undergoes maintenance
6. Insert a new meter history record when the vehicle odometer is updated such as after rental return or inspection

#### **Data Update/Deletion**

1. Update vehicle details such as availability status, mileage or maintenance status
2. Update or replace vehicle photos
3. Modify the pricing rate
4. Deactivate a vehicle from the system without removing historical booking data
5. Update the renewal reminder status, due date and note when the expiry date is approaching or when a reminder has been processed
6. Delete a renewal reminder record if it is entered incorrectly or no longer required
7. Update service history details if corrections are required
8. Update previous odometer information automatically when a new meter update record is added
9. Delete a service history record if it was entered incorrectly without affecting other service records

#### **Data Queries**

1. Retrieve a list of available vehicles based on date and time
2. Retrieve vehicle details and associated photos for display
3. Retrieve applicable rental duration pricing for a selected vehicle
4. Check vehicle availability before confirming a booking
5. List vehicles by status (available, unavailable, under maintenance)
6. Retrieve vehicle usage data for reporting and analysis
7. Retrieve renewal reminders that are approaching the due date
8. Retrieve renewal reminders based by service type and based on vehicle plate and reminder status
9. Retrieve the complete service history of a vehicle based on vehicle plate number
10. Retrieve meter update records to track odometer changes over time

### 3.2.2.3 Local Conceptual Data Model (ERD)

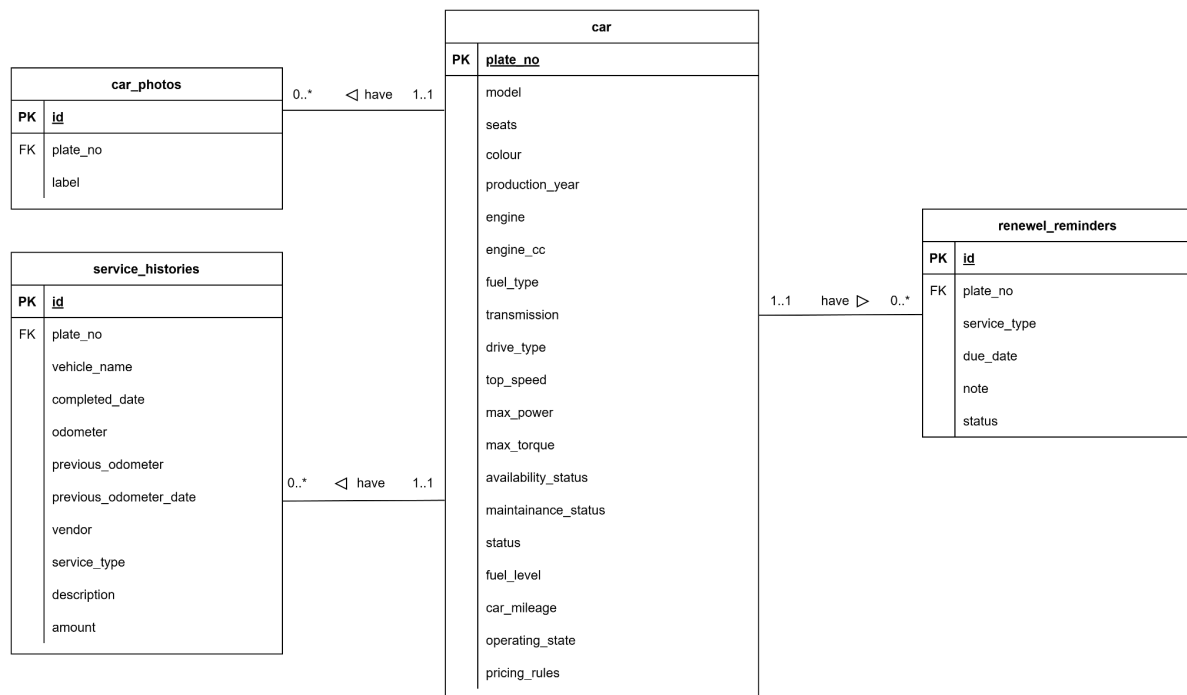


Figure 3.2.2.3.1 Local Conceptual Data Model for Fleet Management Module

### 3.2.3 Booking Management Module

The Booking Module manages the complete vehicle rental lifecycle, from reservation creation to vehicle pick up and return. It coordinates booking details, rental periods, vehicle assignment and booking status updates. This module ensures that vehicles are allocated correctly, rental rules are enforced and booking records are maintained for operational and reporting purposes. It allows customers to create, manage and track vehicle bookings while allowing admin and manager to verify, approve and monitor booking activities.

#### 3.2.3.1 Data Requirements

##### Booking

Each rental is uniquely identified by a rental ID. Stored booking data includes rental ID, booking status, withdrawal receipt, withdrawal status, destination, invoice, payment status, fuel penalty and late penalty. Booking status indicates the current stage of the rental process such as pending, verified, renting, done and rejected. Each rental has one payment and one voucher. Each booking is associated with one vehicle and a customer can make multiple bookings.

##### Pick Up

Pick up records capture information related to vehicle handover to the customer. Each pick up record is uniquely identified by a pick up ID and linked to a rental. Stored data includes pick up ID, rental start date and time, pick up location, vehicle condition images at pick up, dashboard image, damage image if any damage exists on vehicle before pick up, description

about damage part, fuel level at pick up, agreement form, acknowledgement status and pick up report. This information ensures proper documentation of the vehicle condition before the rental begins.

## **Return**

Return records store information related to vehicle return at the end of rental period. Each return record is uniquely identified by a return ID and linked to a rental. Stored data includes return ID, rental end date and time, return location, vehicle condition images at return, dashboard image, damage image if any damage exists in the rental period, damage remarks, fuel level at return, service rating, vehicle condition rating, feedback, acknowledgement of the images for different views and return report.

### **3.2.3.2 Transaction Requirements**

#### **Data Entry**

1. Create a new booking by selecting a vehicle, rental period and destination
2. Record vehicle pickup details during handover
3. Record vehicle return details upon rental completion

#### **Data Update/Deletion**

1. Update booking status throughout the rental lifecycle
2. Update late penalty if the customer returns the vehicle late
3. Update fuel penalty when the return fuel level is less than pickup fuel level
4. Cancel or close bookings without deleting historical records

#### **Data Queries**

1. Retrieve booking details for a specific customer or vehicle
2. Check vehicle availability for a selected rental period
3. Retrieve pickup and return records linked to a booking
4. List verified, in progress, completed or cancelled bookings
5. Retrieve booking history for customer dashboards

### 3.2.3.3 Local Conceptual Data Model (ERD)

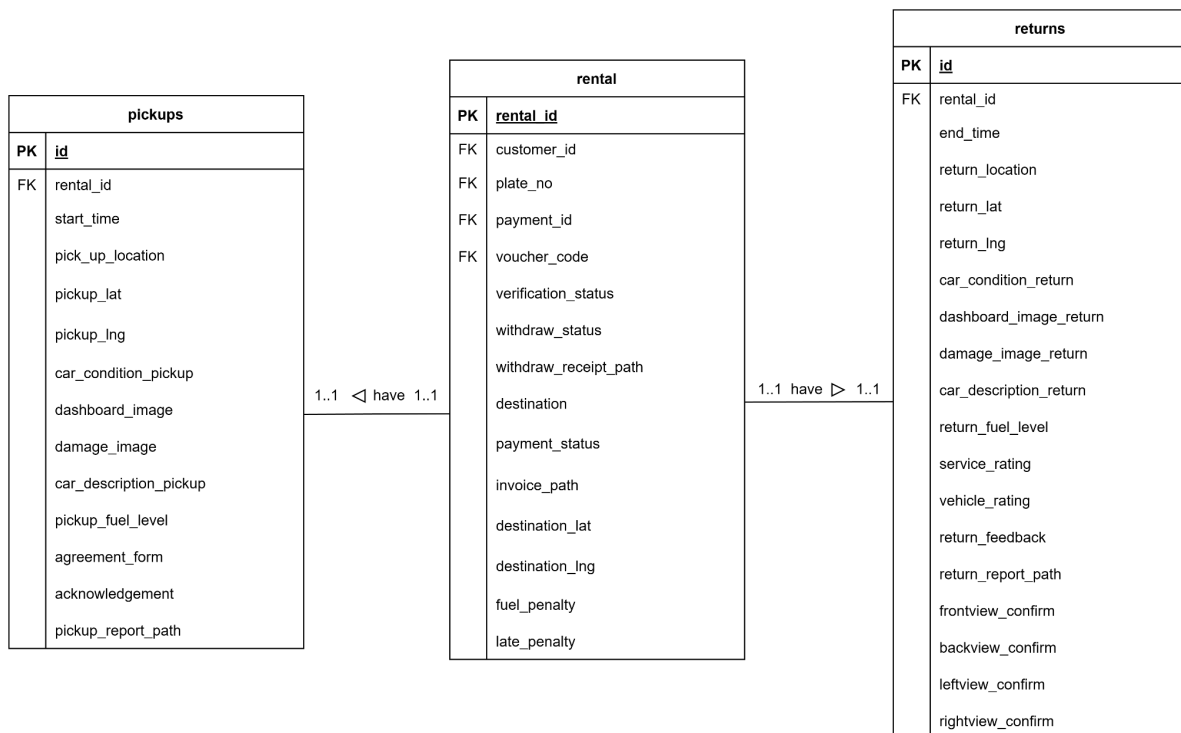


Figure 3.2.3.3.1 Local Conceptual Data Model for Booking Management Module

### 3.2.4 Payment Management Module

This module manages all financial transactions within the Hasta Travel & Tours Vehicle Rental Management System. It handles rental payments, deposits, penalties and refunds while supporting manual payment verification by administrators or managers. It ensures that all payments are properly recorded, verified and linked to the relevant booking and customer records.

#### 3.2.4.1 Data Requirements

Each payment transaction is uniquely identified by a payment ID. Stored data includes payment ID, payment amount, deposit amount, penalty amount, payment type, payment method, receipt image and refund amount.

The payment type indicates whether the transaction is for a booking payment or a penalty charge. The payment method is QR payment. The verification status reflects whether the payment is pending, approved or rejected. Each payment record is linked to the corresponding customer and verified by an authorized admin or manager.

#### 3.2.4.2 Transaction Requirements

##### Data Entry

1. Record a new payment for a booking or penalty
2. Upload payment receipt image for verification
3. Record deposit amount associated with a booking
4. Assign verifying admin/manager during payment verification

## Data Update/Deletion

1. Update payment verification status whether approved or rejected
2. Update deposit balance after successful verification

## Data Queries

1. Retrieve payment details for a specific booking or customer
2. Retrieve pending payments awaiting verification
3. Retrieve verified payments processed by a specific admin or manager
4. Retrieve deposit and penalty payment records
5. Check payment verification status before confirming booking
6. Generate payment summaries for reporting and auditing purposes

### 3.2.4.3 Local Conceptual Data Model (ERD)

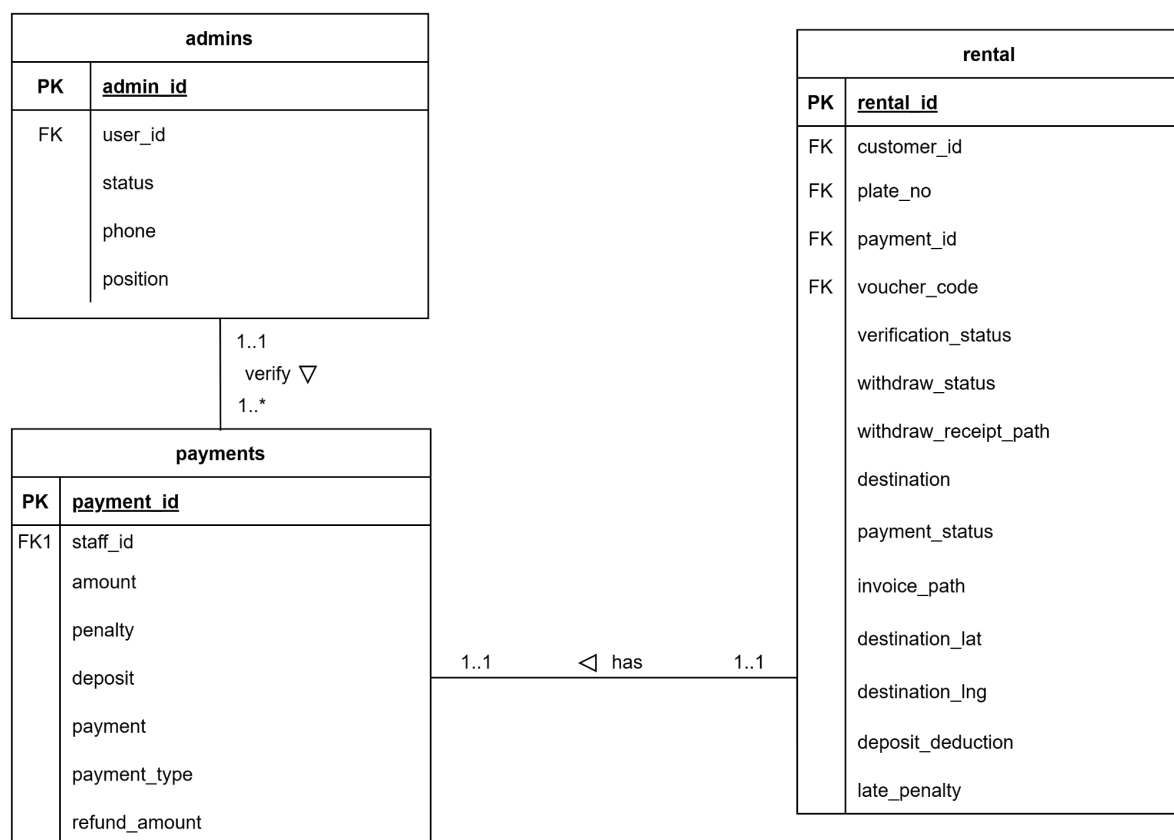


Figure 3.2.4.3.1 Local Conceptual Data Model for Payment Management Module

## 3.2.5 Loyalty and Rewards Module

Loyalty and Rewards Module manages customer incentive programs within the Hasta Travel & Tours Vehicle Rental Management System. It tracks customer rental activity, awards loyalty stamps and manages reward vouchers. This module aims to encourage repeat usage by rewarding loyal customers with discounts and free rental benefits.

### **3.2.5.1 Data Requirements**

#### **Loyalty**

For the loyalty record, it is uniquely identified by a loyalty ID and is linked to a customer. Stored data includes loyalty ID, loyalty type, stamp description, rental counter, number of stamps earned.

The loyalty type indicates the nature of the reward activity such as rental-based stamps. The rental counter would track the number of qualifying rentals while the stamp count reflects accumulated loyalty progress. Loyalty records are generated automatically based on completed bookings that meet eligibility criteria or when the customers redeem the voucher.

#### **Voucher**

Each voucher is uniquely identified by a voucher ID and is linked to a customer. Stored data includes voucher ID, voucher code, voucher description, discount percentage, free rental hours, expiry date and voucher status.

Vouchers are not issued automatically. Customers may manually redeem vouchers through the system once they have sufficient loyalty stamps. Moreover, Admin and Manager can issue the vouchers as part of promotions or special cases. Voucher status indicates whether a voucher is active, used or expired.

### **3.2.5.2 Transaction Requirements**

#### **Data Entry**

1. Generate loyalty records when a qualifying rental is completed or when a customer redeems the voucher
2. Create voucher records through customer redemption
3. Create vouchers manually by Admin or Manager to specific customers
4. Assign voucher codes and reward attributes

#### **Data Update/Deletion**

1. Update loyalty stamp count after eligible bookings
2. Increment rental counter after eligible bookings
3. Deduct rental counter upon successful redemption
4. Update voucher description, expiry date or status by Admin or Manager
5. Deactivate vouchers without deleting historical records

#### **Data Queries**

1. Retrieve loyalty stamp progress for a specific customer
2. Retrieve active, used and expired vouchers for a customer
3. Validate voucher eligibility during booking
4. Retrieve vouchers created or managed by Admin or Manager
5. Generate loyalty and voucher usage for reporting purposes

### 3.2.5.3 Local Conceptual Data Model (ERD)

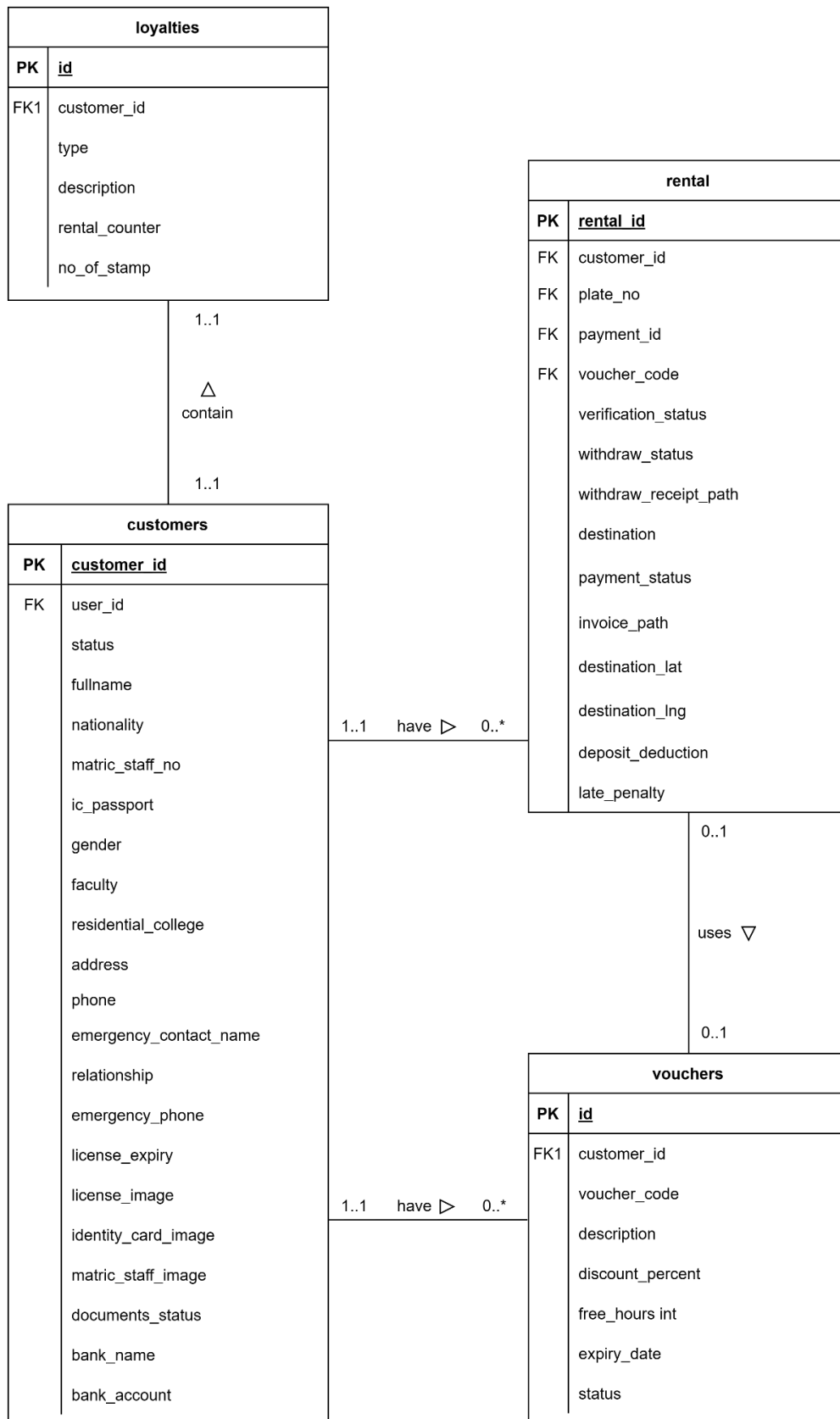


Figure 3.2.5.3.1 Local Conceptual Data Model for Loyalty and Rewards Module

### **3.2.6 Report and Analytics Module**

The Report and Analytics Module provides administrative and managerial insights by analyzing and summarizing data collected from existing system modules. It retrieves and processes data from the User, Vehicle, Booking and Payment modules to generate meaningful reports and statistical summaries for operational monitoring and decision-making purposes.

#### **3.2.6.1 Data Requirements**

##### **Sales and Revenue Data**

Sales and revenue information is obtained from verified payment records in the Payment Module, including payment amount, deposit value, payment type and transaction timestamps. These records allow revenue to be aggregated and analyzed according to different time periods such as daily, monthly and yearly summaries. Moreover, it can be categorized based on vehicle details and customer types retrieved from the Fleet and Use Modules.

##### **Fleet Performance**

Fleet performance data is generated by analyzing vehicle availability and booking information stored in the Fleet and Booking Modules. Maintenance performance and downtime are assessed using vehicle maintenance and availability status fields. Meanwhile for the vehicle popularity metrics, they are derived from booking frequency data to identify high-demand models within the fleet.

##### **Customer Feedback**

Customer feedback data is sourced from feedback records submitted after booking completion. This includes numerical service and vehicle ratings, along with textual feedback provided by customers. Feedback data is retrieved to support trend analysis and service quality evaluation.

#### **3.2.6.2 Transaction Requirements**

##### **Data Entry**

1. Generate monthly and yearly sales reports using verified payment data
2. Generate fleet utilization analysis comparing booked hours against total available hours
3. Generate customer feedback summaries after rental completion

##### **Data Update/Deletion**

1. Regenerate reports dynamically based on updated criteria
2. Update report parameters such as data range filters to regenerate views dynamically

##### **Data Queries**

1. Retrieve total revenue for a specific date range
2. Retrieve the average vehicle rating scores
3. List all feedback entries with low ratings which is less than 3 stars for manager review
4. Export report data in PDF and Excel format for offline analysis

## **4.0 Database conceptual design**

### **4.1 Conceptual ERD**

Figure 4.1.1 is an example of how Hasta Travel Vehicle Management System manages information within its systems to create a data structure that is accurate and efficient by maintaining an accurate representation of the information stored on the system. Some of the basic concepts of databases have been used within this design such as entity specialization. For example, a user can act as either a customer or an admin. Defined cardinalities were created to define the relationships of connections between data elements. For instance, the customer to the rental relationship has a cardinality of 1:M. Each Rental is only associated with one vehicle, one pickup record and one return record and a payment reference to maintain transactional consistency. In addition, all of the less commonly used Supporting Entities like loyalties, payments, vehicle pictures, service records and renewal reminders have been separated into their own tables to allow for data normalization and to limit data redundancy. This ERD provides a structured, scalable model of a Vehicle Rental Company's operations in the real world and also provides for future expansion of the system.

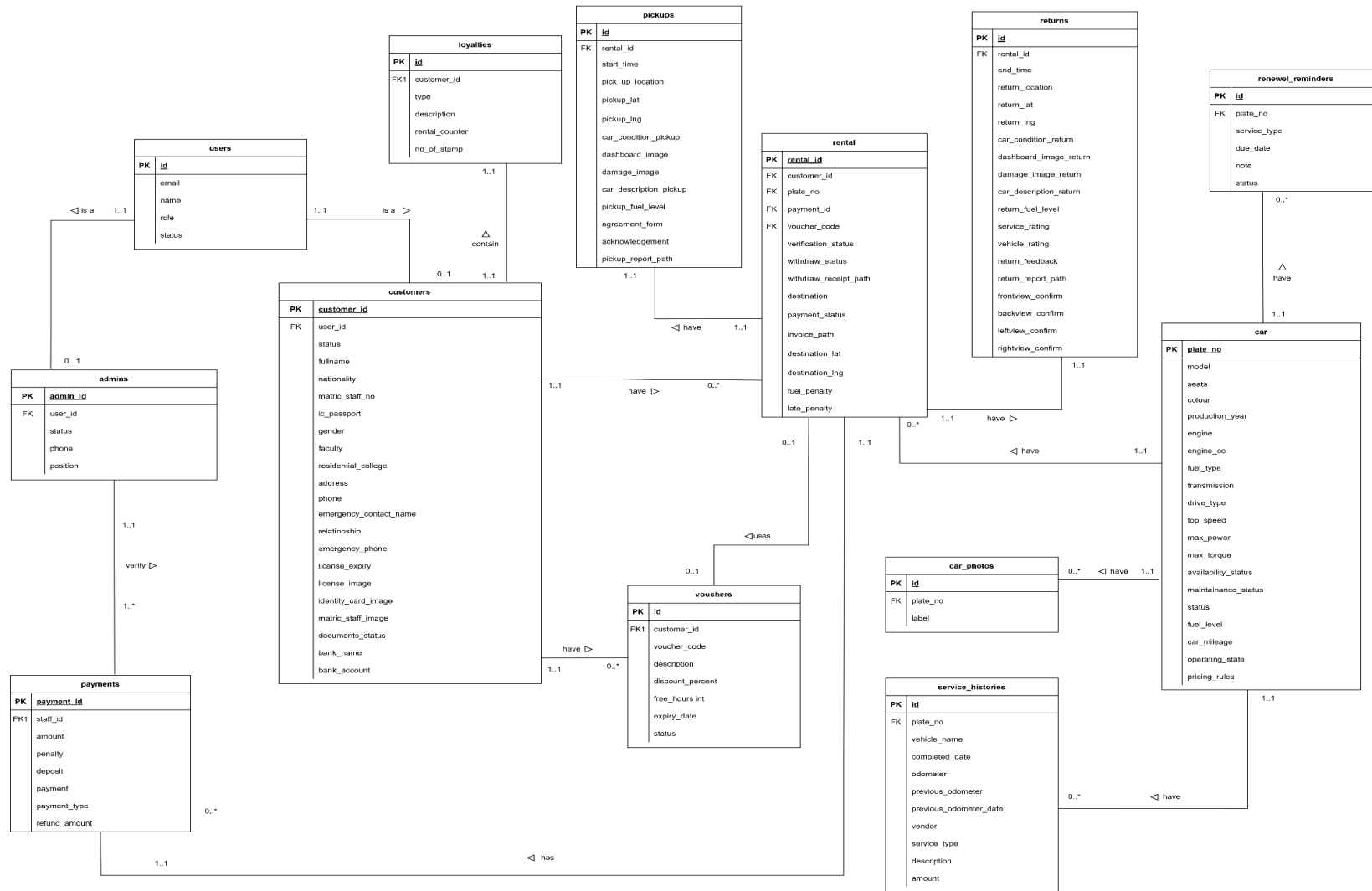


Figure 4.1.1 Entity Relationship Diagram (ERD)

## **4.2 Enhanced ERD (EERD)**

Figure 4.2.1 builds upon the information captured in the basic ERD to provide a means of representing more advanced database concepts such as specialization, mandatory participation, and enhanced relationships. The EERD includes an example of a mandatory OR (ISA) specialization, where every user must be classified either as a customer or an admin, thus enforcing role integrity at the database level. The rental entity is identified as the primary transaction and is mandatorily related to both pickup and return entities to provide for a complete rental lifecycle. The additional entities such as vouchers that have a voucher redemption relationship, loyalties, payments, car photos, service histories, and renewal reminders, have been separated to allow for normalization of data within the database and to avoid redundant data. The overall result of the EERD is a more expressive and accurate database design that captures business rules, business constraints, and business logic more accurately than a basic ERD.

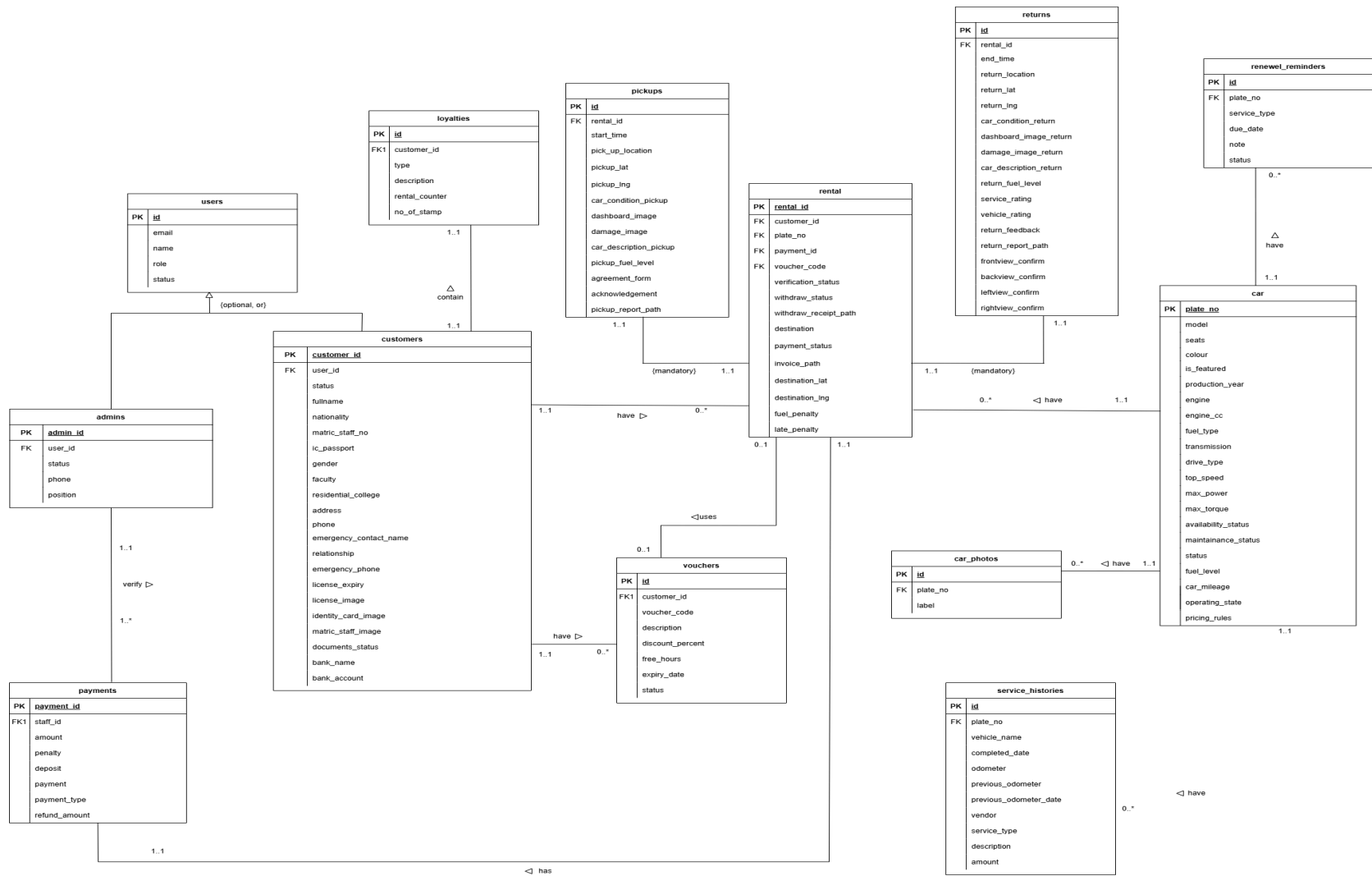


Figure 4.2.1 Enhanced Entity Relationship Diagram (EERD)

## 5.0 Data Dictionary

### 1. admins

| Column   | Data Type | Length | Description                         | Allowed Values     | Relationship           | Required |
|----------|-----------|--------|-------------------------------------|--------------------|------------------------|----------|
| admin_id | varchar   | 50     | Unique admin identifier             | -                  | Primary Key            | Yes      |
| user_id  | bigint    | -      | Unique user identifier              | -                  | Foreign Key => user.id | Yes      |
| status   | varchar   | 20     | Current status of the admin account | active/deactivated | -                      | Yes      |
| phone    | varchar   | 20     | Admin's contact phone number        | -                  | -                      | No       |
| position | varchar   | 20     | Admin's job position                | -                  | -                      | No       |

### 2. car

| Column   | Data Type | Length | Description            | Allowed Values | Relationship | Required |
|----------|-----------|--------|------------------------|----------------|--------------|----------|
| plate_no | varchar   | 20     | Vehicle's plate number | Alphanumeric   | Primary Key  | Yes      |
| model    | varchar   | 150    | Vehicle's model        | -              | -            | Yes      |
| seats    | tinyint   | -      | Number of seats        | 1 – 9          | -            | No       |

|                     |         |       |                                |                         |   |     |
|---------------------|---------|-------|--------------------------------|-------------------------|---|-----|
| colour              | varchar | 30    | Vehicle colour                 | -                       | - | No  |
| is_featured         | tinyint | 1     | Featured vehicle flag          | 0 = No, 1 = Yes         | - | Yes |
| availability_status | tinyint | 1     | Availability of vehicle        | available / unavailable | - | Yes |
| status              | varchar | 20    | Operational status of car      | active / retired        | - | Yes |
| fuel_level          | integer | -     | The fuel level for the vehicle | -                       | - | Yes |
| car_mileage         | decimal | 10, 2 | Vehicle's mileage              | $\geq 0$                | - | Yes |
| production_year     | year    | -     | Year of manufacture            | YYYY                    | - | No  |
| engine              | varchar | 100   | Engine description             | -                       | - | No  |
| engine_cc           | integer | -     | Engine capacity(cc)            | -                       | - | No  |
| fuel_type           | varchar | 20    | Type of fuel                   | Petrol/Diesel/EV        | - | No  |
| transmission        | varchar | 20    | Transmission type              | Automatic / Manual      | - | No  |
| drive_type          | varchar | 20    | Drive type                     | FWD/RWD/AWD             | - | No  |
| top_speed           | integer | -     | Maximum speed (km/h)           | -                       | - | No  |

|                    |         |      |                                     |                               |   |     |
|--------------------|---------|------|-------------------------------------|-------------------------------|---|-----|
| max_power          | decimal | 8, 2 | Maximum power (HP)                  | -                             | - | No  |
| max_torque         | decimal | 8, 2 | Maximum torque (Nm)                 | -                             | - | No  |
| maintenance_status | tinyint | 1    | Maintenance state                   | 0 = OK, 1 = Under Maintenance | - | Yes |
| operating_state    | varchar | 50   | State where vehicle operates        | -                             | - | No  |
| pricing_rules      | json    | -    | Hourly pricing rates of the vehicle | -                             | - | Yes |

### 3. car\_photos

| Column   | Data Type         | Length | Description                                     | Allowed Values                        | Relationship                | Required |
|----------|-------------------|--------|-------------------------------------------------|---------------------------------------|-----------------------------|----------|
| id       | bigint (UNSIGNED) | -      | Unique identifier for each vehicle photo record | Auto-increment integer                | Primary Key                 | Yes      |
| plate_no | varchar           | 20     | Vehicle's plate number                          | Alphanumeric                          | Foreign key => car.plate_no | Yes      |
| label    | varchar           | 50     | Label of the image                              | Front, Back, Left, Right, Interior 1, | -                           | No       |

|  |  |  |  |            |  |  |
|--|--|--|--|------------|--|--|
|  |  |  |  | Interior 2 |  |  |
|--|--|--|--|------------|--|--|

4. customers

| Column          | Data Type | Length | Description                                    | Allowed Values                         | Relationship              | Required |
|-----------------|-----------|--------|------------------------------------------------|----------------------------------------|---------------------------|----------|
| customer_id     | varchar   | 50     | Unique customer identifier                     | -                                      | Primary key               | Yes      |
| user_id         | bigint    | -      | Unique user identifier                         | -                                      | Foreign Key =><br>user.id | Yes      |
| status          | varchar   | 20     | Current status of the customer account         | active/<br>blacklisted                 | -                         | Yes      |
| fullname        | varchar   | 100    | Customer's full name as per IC/passport        | -                                      | -                         | No       |
| nationality     | varchar   | 100    | Customer's nationality                         | -                                      | -                         | No       |
| matric_staff_no | varchar   | 20     | Student or staff matric number                 | -                                      | Unique key                | No       |
| ic_passport     | varchar   | 15     | IC number (local) or passport number (foreign) | -                                      | -                         | No       |
| gender          | varchar   | 50     | Customer's gender                              | "Male",<br>"Female",<br>"Prefer not to | -                         | No       |

|                            |         |     |                                                                  |                        |            |    |
|----------------------------|---------|-----|------------------------------------------------------------------|------------------------|------------|----|
|                            |         |     |                                                                  | say”                   |            |    |
| faculty                    | varchar | 100 | Customer’s faculty                                               | -                      | -          | No |
| residential_coll<br>ege    | varchar | 100 | Customer’s residential<br>college                                | -                      | -          | No |
| address                    | text    | -   | Customer’s home address                                          | -                      | -          | No |
| phone                      | varchar | 20  | Customer’s contact phone<br>number                               | Digits, ‘+’<br>allowed | Unique key | No |
| emergency_con<br>tact_name | varchar | 100 | Name of emergency contact<br>person                              | -                      | -          | No |
| relationship               | varchar | 50  | Relationship between<br>customer and emergency<br>contact person | -                      | -          | No |
| emergency_pho<br>ne        | varchar | 20  | Emergency contact phone<br>number                                | Digits ‘+’<br>allowed  | -          | No |
| license_expiry             | date    | -   | Driving license expiry date                                      | -                      | -          | No |
| license_image              | varchar | 255 | File path or URL to driving<br>license image                     | -                      | -          | No |
| identity_card_i            | varchar | 255 | File path or URL to identity                                     | -                      | -          | No |

|                    |         |     |                                            |                                         |   |    |
|--------------------|---------|-----|--------------------------------------------|-----------------------------------------|---|----|
| mage               |         |     | card image                                 |                                         |   |    |
| matric_staff_image | varchar | 255 | File path or URL to matric card image      | -                                       | - | No |
| documents_status   | enum    | -   | Verification status of submitted documents | 'pending',<br>'approved',<br>'rejected' |   | No |
| bank_name          | varchar | 100 | Customer's bank name                       | -                                       | - | No |
| bank_account       | varchar | 30  | Customer's bank account number             | -                                       | - | No |

#### 5. loyalties

| Column      | Data Type | Length | Description                                     | Allowed Values          | Relationship                           | Required |
|-------------|-----------|--------|-------------------------------------------------|-------------------------|----------------------------------------|----------|
| id          | bigint    | -      | Unique loyalty identifier                       | -                       | Primary Key                            | Yes      |
| customer_id | varchar   | 50     | ID of the customer who owns this loyalty record | -                       | Foreign Key -><br>customer.customer_id | Yes      |
| type        | varchar   | 50     | Indicates if stamp was earned or redeemed       | 'earned',<br>'redeemed' | -                                      | Yes      |

|                |         |     |                                   |            |   |     |
|----------------|---------|-----|-----------------------------------|------------|---|-----|
| description    | varchar | 255 | Description of the loyalty action | -          | - | Yes |
| rental_counter | integer | -   | Current Redeemable Stamps         | 0 or above | - | Yes |
| no_of_stamp    | integer | -   | Lifetime Total Stamps             | 0 or above | - | Yes |

6. payments

| Column       | Data Type | Length | Description                      | Allowed Values | Relationship                   | Required |
|--------------|-----------|--------|----------------------------------|----------------|--------------------------------|----------|
| payment_id   | varchar   | 20     | Unique payment identifier        | -              | Primary Key                    | Yes      |
| staff_id     | bigint    | 50     | Unique staff identifier          | -              | Foreign Key -> admins.admin_id | No       |
| amount       | decimal   | 10, 2  | Amount need to pay               | -              | -                              | Yes      |
| penalty      | decimal   | 10, 2  | Penalty amount that needs to pay | -              | -                              | Yes      |
| deposit      | decimal   | 10, 2  | Deposit amount that needs to pay | -              | -                              | Yes      |
| payment      | varchar   | 20     | Payment method to choose         | Valid bank QR  | -                              | Yes      |
| payment_type | varchar   | 20     | Reason of payment whether        | booking /      | -                              | Yes      |

|               |         |       |                                                       |         |   |     |
|---------------|---------|-------|-------------------------------------------------------|---------|---|-----|
|               |         |       | for rental or penalty                                 | penalty |   |     |
| refund_amount | decimal | 10, 2 | The amount that needed to be refunded to the customer | -       | - | Yes |

7. renewal\_reminders

| Column       | Data Type         | Length | Description                                        | Allowed Values                                                       | Relationship                | Required |
|--------------|-------------------|--------|----------------------------------------------------|----------------------------------------------------------------------|-----------------------------|----------|
| id           | bigint (UNSIGNED) | -      | Unique identifier for each renewal reminder record | Auto-increment integer                                               | Primary Key                 | Yes      |
| plate_no     | varchar           | 20     | Vehicle plate number associated with the reminder  | Alphanumeric                                                         | Foreign key => car.plate_no | Yes      |
| service_type | enum              | -      | Type of renewal or service reminder                | Insurance, Investor Commission, Monthly Installment, Roadtax, Others | -                           | Yes      |
| due_date     | date              | -      | Due date for the renewal or service                | Valid date                                                           | -                           | Yes      |
| note         | text              | -      | Additional remarks or notes for the                | Free text                                                            | -                           |          |

|        |      |   |                         |                               |   |     |
|--------|------|---|-------------------------|-------------------------------|---|-----|
|        |      |   | reminder                |                               |   |     |
| status | enum | - | Current reminder status | upcoming, notified, completed | - | Yes |

8. rental

| Column       | Data Type | Length | Description                                    | Allowed Values                   | Relationship                         | Required |
|--------------|-----------|--------|------------------------------------------------|----------------------------------|--------------------------------------|----------|
| rental_id    | bigint    | -      | Unique identifier for each rental transaction  | Auto-increment positive integer  | Primary Key                          | Yes      |
| customer_id  | varchar   | 50     | Identifier of the customer who made the rental | Existing customer IDs            | Foreign Key -> customer.customer_id  | Yes      |
| plate_no     | varchar   | 10     | Vehicle plate number rented                    | Registered vehicle plate numbers | Foreign Key -> car.plate_no          | Yes      |
| payment_id   | varchar   | 50     | Unique payment id                              | System-generated payment IDs     | Foreign Key -> payment.payment_id    | Yes      |
| voucher_code | varchar   | 50     | Voucher code applied to rental                 | Existing voucher                 | Foreign Key -> vouchers.voucher_code | No       |

|                       |         |      |                                                             |                                                                    |    |     |
|-----------------------|---------|------|-------------------------------------------------------------|--------------------------------------------------------------------|----|-----|
|                       |         |      |                                                             | codes                                                              | de |     |
| verification_status   | varchar | 50   | Status of verification                                      | pending/<br>verified/<br>done/<br>renting/<br>Cancelled/<br>failed | -  | Yes |
| withdraw_status       | varchar | 20   | Status of deposit<br>withdrawal/refund                      | pending/<br>done                                                   | -  | Yes |
| withdraw_receipt_path | varchar | 255  | The receipt for<br>withdraw reposit that<br>upload by admin | Valid file<br>path                                                 | -  | No  |
| destination           | varchar | 255  | Destination entered by<br>customer                          | Free text<br>location                                              | -  | Yes |
| payment_status        | varchar | 50   | The status to record the<br>payment of user                 | pending/<br>paid/<br>failed/<br>cancelled                          | -  | No  |
| invoice_path          | varchar | 255  | The autogenerated<br>invoice of this rental                 | Valid file<br>path                                                 | -  | No  |
| destination_lat       | decimal | 10,7 | Latitude of destination<br>location                         | Valid latitude<br>coordinate                                       | -  | Yes |

|                 |         |      |                                                                    |                            |   |     |
|-----------------|---------|------|--------------------------------------------------------------------|----------------------------|---|-----|
| destination_lng | decimal | 10,7 | Longitude of destination location                                  | Valid longitude coordinate | - | Yes |
| fuel_penalty    | decimal | 8, 2 | Charges incurred when fuel level during return is less than pickup | -                          | - |     |
| late_penalty    | decimal | 10,2 | Charges incurred due to late return                                | -                          | - | Yes |

9. pickups

| Column           | Data Type | Length | Description                                       | Allowed Values            | Relationship                   | Required |
|------------------|-----------|--------|---------------------------------------------------|---------------------------|--------------------------------|----------|
| id               | bigint    | -      | Unique booking identifier                         | Auto-increment            | Primary Key                    | Yes      |
| rental_id        | bigint    | -      | References the rental associated with this pickup | Existing rental.rental_id | Foreign Key-> rental.rental_id | Yes      |
| start_time       | datetime  | -      | Scheduled pickup start time and date              | Valid datetime            | -                              | Yes      |
| pick_up_location | varchar   | 255    | The pickup location                               | Text                      | -                              | Yes      |
| pickup_lat       | decimal   | 10,7   | Latitude coordinate of pickup location            | Any valid latitude        | -                              | No       |

|                        |         |      |                                                         |                     |   |     |
|------------------------|---------|------|---------------------------------------------------------|---------------------|---|-----|
| pickup_lng             | decimal | 10,7 | Longitude coordinate of pickup location                 | Any valid longitude | - | No  |
| car_condition_pickup   | text    | -    | Text describing vehicle condition at pickup             | Text                | - | No  |
| dashboard_image        | varchar | 255  | Image path of vehicle dashboard at pickup               | File path           | - | No  |
| damage_image           | varchar | 255  | Image path showing any vehicle damage at pickup         | File path           | - | No  |
| car_description_pickup | text    | -    | Additional notes describing vehicle condition at pickup | Text                | - | No  |
| pickup_fuel_level      | integer | 1    | Fuel level at pickup                                    | 1 to 8              | - | No  |
| agreement_form         | varchar | 255  | Signed rental agreement file path                       | File path           | - | No  |
| acknowledgement        | tinyint | 1    | Customer acknowledgement of pickup                      | 0,1                 | - | Yes |
| pickup_report_path     | varchar | 255  | Generated pickup inspection report file path            | File path           | - | No  |

10. returns

| Column | Data Type | Length | Description | Allowed | Relationship | Required |
|--------|-----------|--------|-------------|---------|--------------|----------|
|--------|-----------|--------|-------------|---------|--------------|----------|

|                        |          |      |                                                   | Values                    |                                     |     |
|------------------------|----------|------|---------------------------------------------------|---------------------------|-------------------------------------|-----|
| id                     | bigint   | -    | Unique identifier for return record               | Auto-increment            | Primary Key                         | Yes |
| rental_id              | bigint   | -    | References the rental associated with this return | Existing rental.rental_id | Foreign Key -> customer.customer_id | Yes |
| end_time               | datetime | -    | Actual return date and time of the vehicle        | Valid datetime            | -                                   | Yes |
| return_location        | varchar  | 255  | Vehicle return location name or address           | Text                      | -                                   | Yes |
| return_lat             | decimal  | 10,7 | Latitude of return location                       | Any valid latitude        | -                                   | No  |
| return_lng             | decimal  | 10,7 | Longitude of return location                      | Any valid longitude       | -                                   | No  |
| car_condition_return   | text     | -    | Text describing vehicle condition at return       | Text                      | -                                   | No  |
| dashboard_image_return | varchar  | 255  | Image path of dashboard at return                 | File path                 | -                                   | No  |
| dashboard_image_return | varchar  | 255  | Image path showing vehicle damage at return       | File path                 | -                                   | No  |

|                        |         |     |                                                      |           |   |     |
|------------------------|---------|-----|------------------------------------------------------|-----------|---|-----|
| car_description_return | text    | -   | Additional notes on vehicle condition during return  | text      | - | No  |
| return_fuel_level      | int     | -   | Fuel level at time of return                         | 0-8       | - | No  |
| service_rating         | int     | -   | Customer rating for rental service                   | 1-5       | - | No  |
| vehicle_rating         | int     | -   | Customer rating for vehicle condition                | 1-5       | - | No  |
| return_feedback        | text    | -   | Customer feedback or comments after return           | text      | - | No  |
| return_report_path     | varchar | 255 | Generated return inspection report file path         | File path | - | No  |
| frontview_confirmation | tinyint | 1   | Confirmation that front view inspection is completed | 0/1       | - | Yes |
| backview_confirmation  | tinyint | 1   | Confirmation that back view inspection is completed  | 0/1       | - | Yes |
| leftview_confirmation  | tinyint | 1   | Confirmation that left view inspection is completed  | 0/1       | - | Yes |
| rightview_confirmation | tinyint | 1   | Confirmation that right view inspection is completed | 0/1       | - | Yes |

11. service\_histories

| Column                 | Data Type         | Length | Description                                      | Allowed Values         | Relationship                | Required |
|------------------------|-------------------|--------|--------------------------------------------------|------------------------|-----------------------------|----------|
| id                     | bigint (UNSIGNED) | -      | Unique identifier for service record             | Auto-increment integer | Primary Key                 | Yes      |
| plate_no               | varchar           | 20     | Vehicle plate number associated with the service | Alphanumeric           | Foreign key => car.plate_no | Yes      |
| vehicle_name           | varchar           | 100    | Vehicle model/name                               | Text                   | -                           | Yes      |
| completed_date         | date              | -      | Date service was completed                       | Valid date             | -                           | Yes      |
| odometer               | integer           | -      | Vehicle mileage at service time (km)             | $\geq 0$               | -                           | No       |
| previous_odometer      | integer           | -      | Previous recorded mileage                        | $\geq 0$               | -                           | No       |
| previous_odometer_date | date              | -      | Date of previous mileage record                  | Valid date             | -                           | No       |

|              |         |      |                             |                                                       |   |     |
|--------------|---------|------|-----------------------------|-------------------------------------------------------|---|-----|
| vendor       | varchar | 100  | Service provider / workshop | Text                                                  | - | No  |
| service_type | varchar | 50   | Type of service             | Oil Change,<br>Meter Update,<br>Repair,<br>Inspection | - | No  |
| description  | text    | -    | Additional service details  | Free text                                             | - | No  |
| amount       | decimal | 10,2 | Service cost (MYR)          | $\geq 0.00$                                           | - | Yes |

12. users

| Column | Data Type | Length | Description                                                            | Allowed Values                 | Relationship | Required |
|--------|-----------|--------|------------------------------------------------------------------------|--------------------------------|--------------|----------|
| id     | varchar   | -      | Unique user identifier used for authentication and internal references | -                              | Primary key  | yes      |
| email  | varchar   | 100    | User's login email address                                             | Valid email format             | Unique key   | yes      |
| role   | varchar   | 50     | Defines the role of the user in the system                             | customer/<br>admin/<br>manager | -            | yes      |
| name   | varchar   | 50     | Username use in system                                                 | -                              | -            | Yes      |

|        |         |    |                |                     |   |     |
|--------|---------|----|----------------|---------------------|---|-----|
| status | varchar | 20 | account status | active / deactivate | - | Yes |
|--------|---------|----|----------------|---------------------|---|-----|

### 13. vouchers

| Column           | Data Type | Length | Description                             | Allowed Values | Relationship                        | Required |
|------------------|-----------|--------|-----------------------------------------|----------------|-------------------------------------|----------|
| id               | bigint    | -      | Unique identifier for the voucher       | -              | Primary key                         | Yes      |
| customer_id      | varchar   | 50     | ID of the customer who owns the voucher | -              | Foreign Key -> customer.customer_id | Yes      |
| voucher_code     | varchar   | 50     | Unique code for each voucher            | -              | Unique Key                          | Yes      |
| description      | string    | 255    | Detail about the voucher                | -              | -                                   | No       |
| discount_percent | integer   | -      | Percentage discount amount              | -              | -                                   | No       |
| free_hours       | integer   | -      | Free rental hours granted               | -              | -                                   | No       |
| expiry_date      | date      | -      | The date when the voucher expires       | -              | -                                   | Yes      |

|        |         |    |                              |                                |   |     |
|--------|---------|----|------------------------------|--------------------------------|---|-----|
| status | varchar | 50 | Current state of the voucher | 'active', 'used',<br>'expired' | - | Yes |
|--------|---------|----|------------------------------|--------------------------------|---|-----|

## 6.0 Summary

The proposed system for Hasta Travel and Tours represents a strategic shift from traditional manual operations to a comprehensive digital platform designed to optimize the vehicle rental booking process. This transformation is essential for unlocking the company's full business potential while eliminating the inefficiencies that currently compromise productivity and service quality. By incorporating advanced automation, intuitive design, and streamlined workflows, the system is positioned to deliver a superior experience for both customers and internal stakeholders.

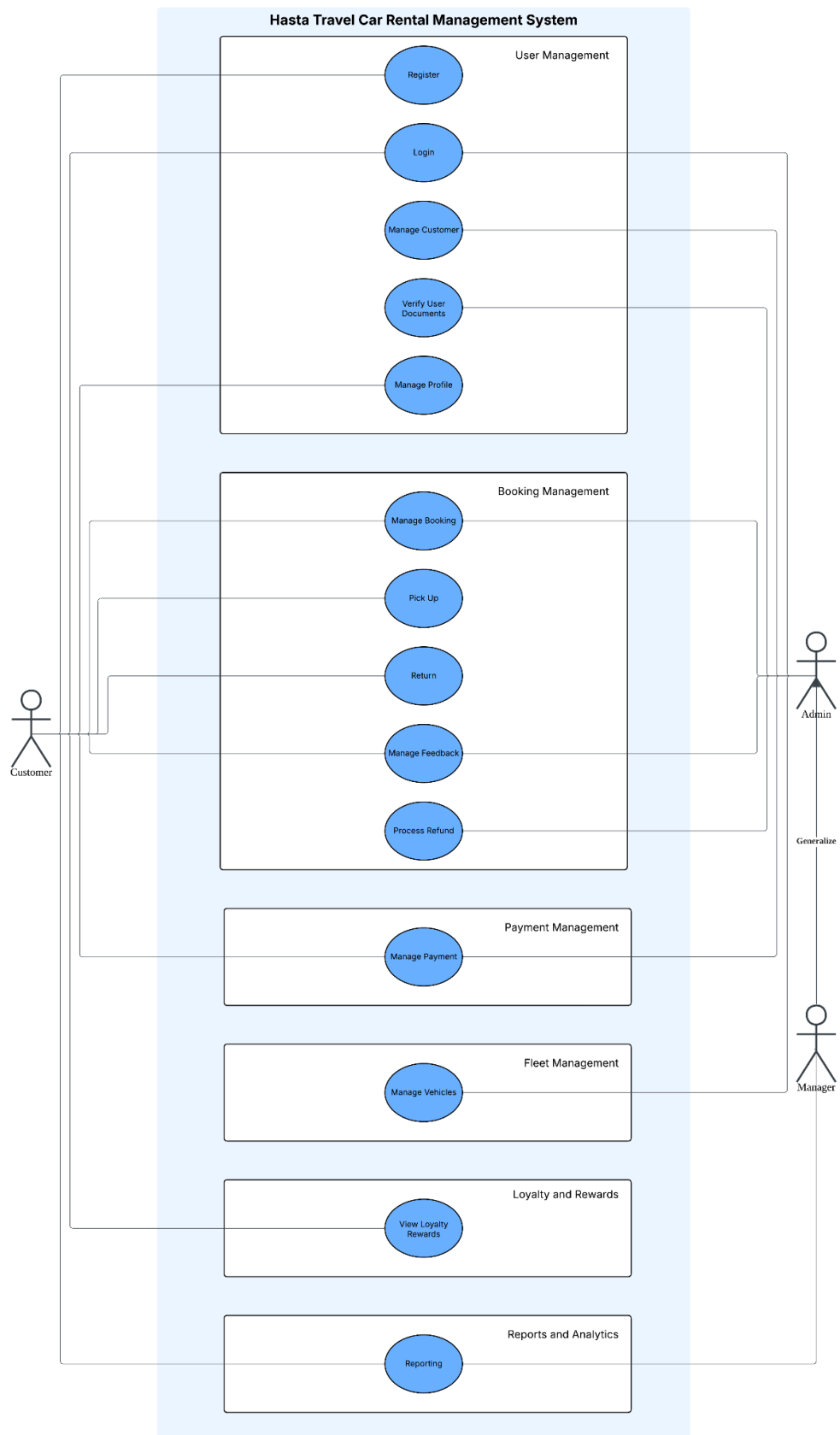
Central to the system are features that prioritize convenience and accessibility. Customers will benefit from a dynamic online booking form that displays real-time vehicle availability and pricing, complemented by automated notifications and instant e-receipt generation. The mobile-responsive interface ensures that users can complete transactions quickly and effortlessly from any device, fostering greater confidence and satisfaction throughout the booking journey.

From an administrative perspective, the system strengthens data accuracy and accountability by automating essential functions such as payment verification, vehicle condition documentation, and revenue tracking. Real-time data synchronization minimizes the likelihood of human error while providing transparent access to financial and rental records. Staff and management gain enhanced operational oversight through organized digital reports and centralized documentation, enabling more informed decision-making.

Ultimately, this system establishes a robust foundation for operational excellence by integrating efficiency, reliability, and transparency into every aspect of the business. The accompanying diagrams and documentation within this proposal provide a detailed blueprint of the database architecture, offering clear insight into the processes, entities, and attributes that underpin the Hasta Travel System.

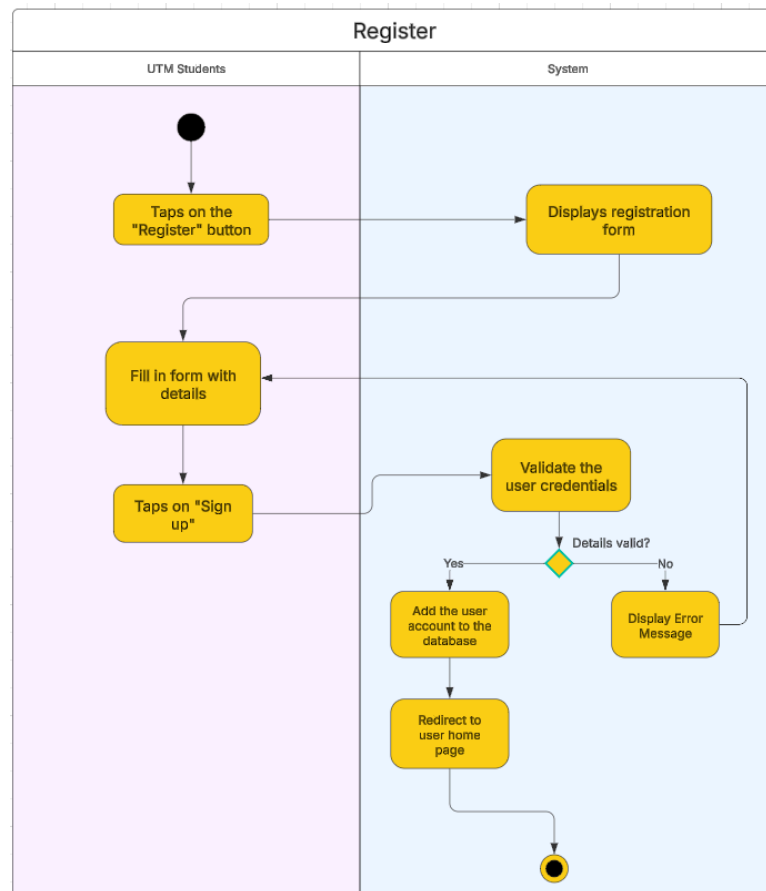
## 7.0 Appendices

### 7.1 Use Case Diagram

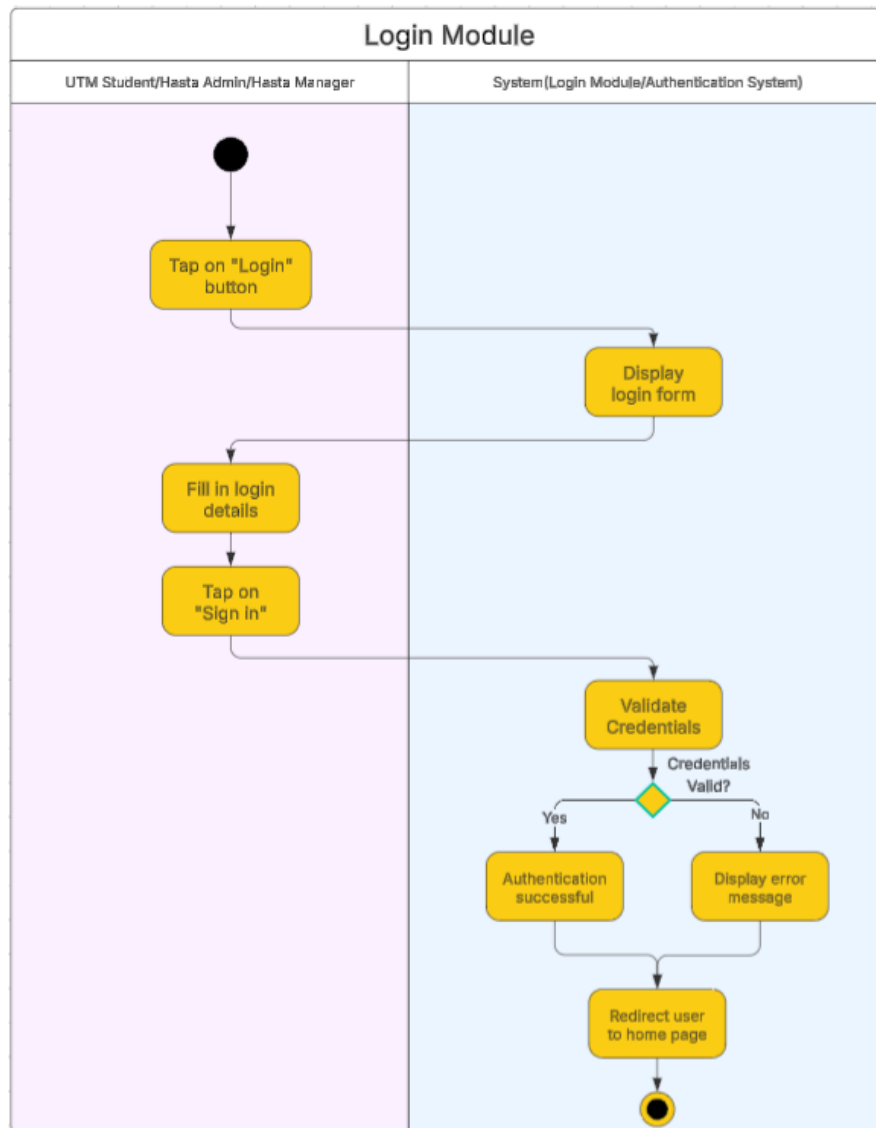


## 7.2 Activity Diagram

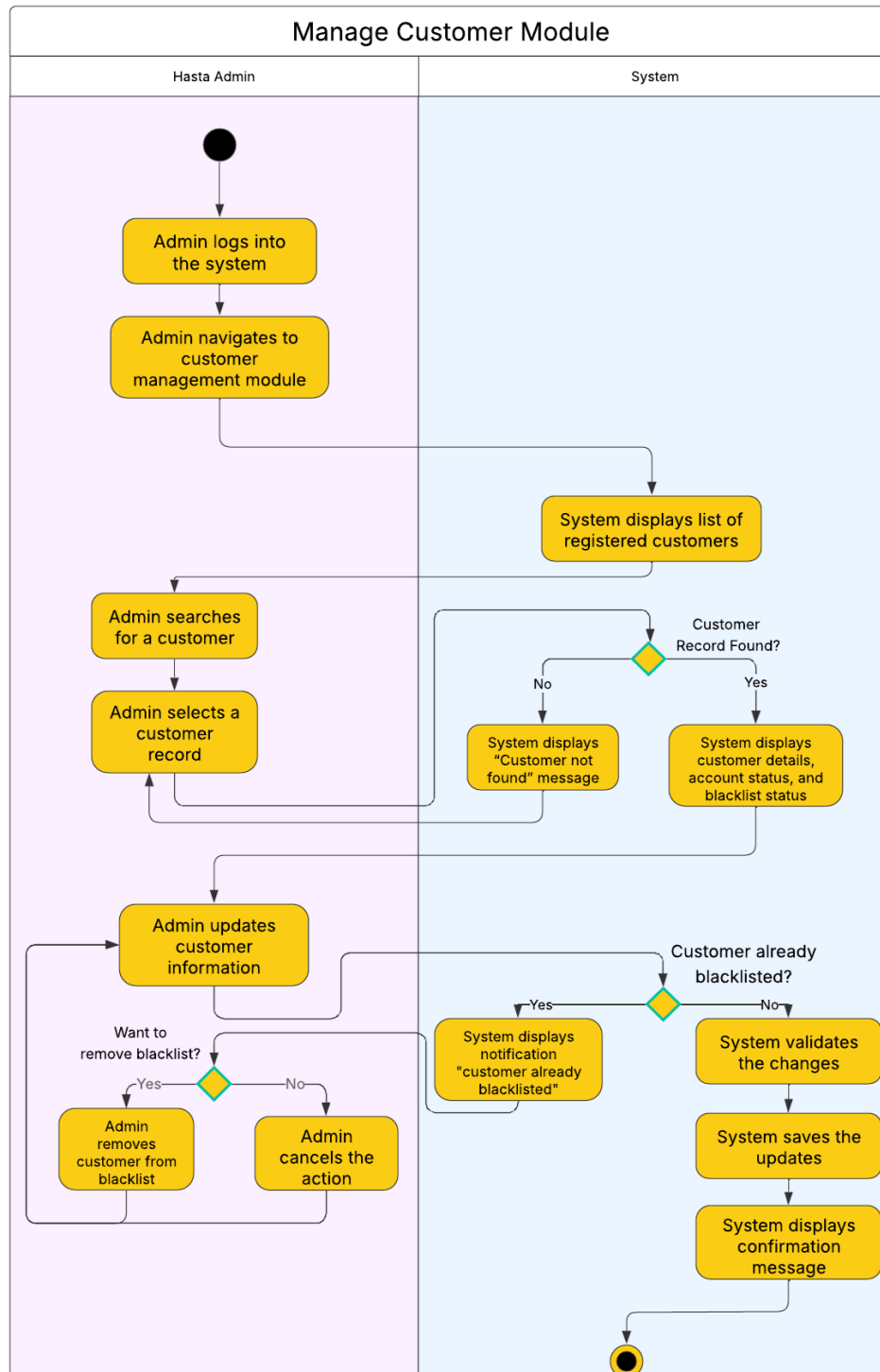
### 1. Register



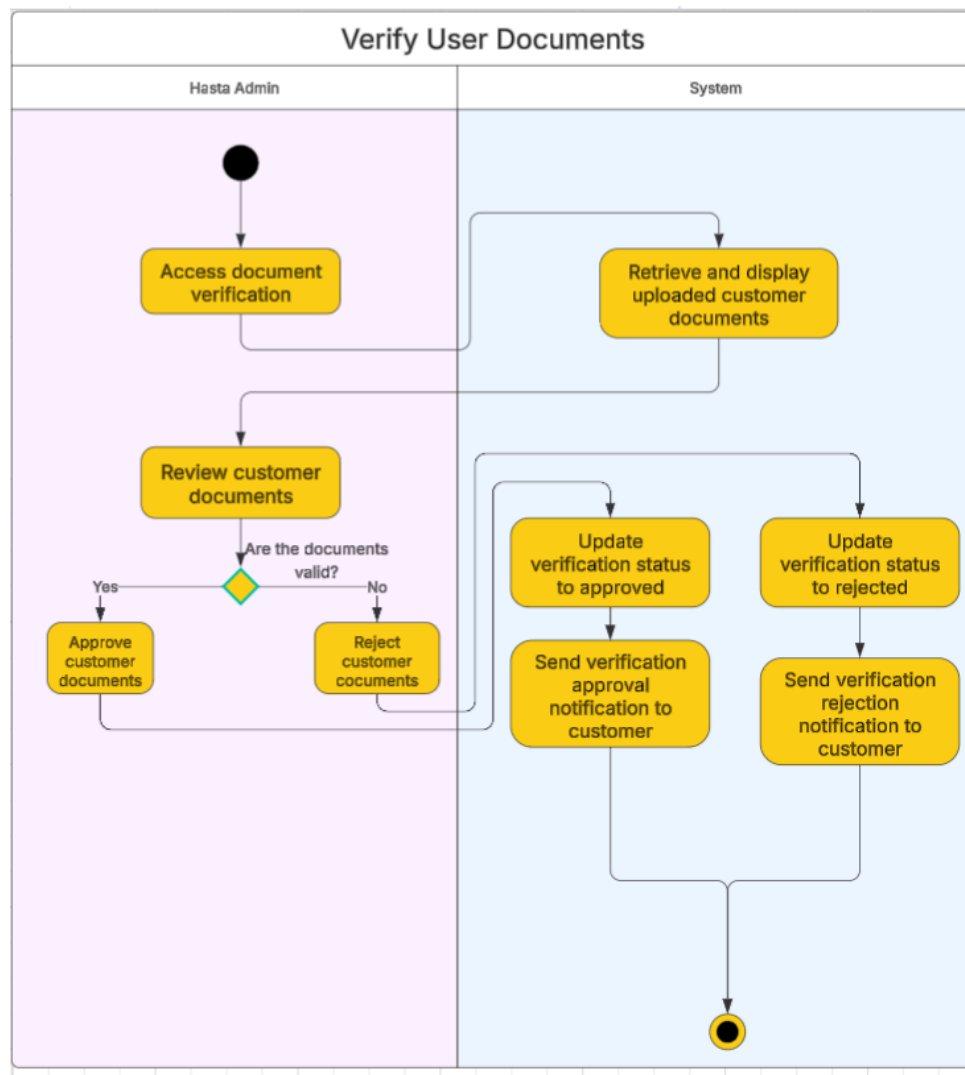
## 2. Login



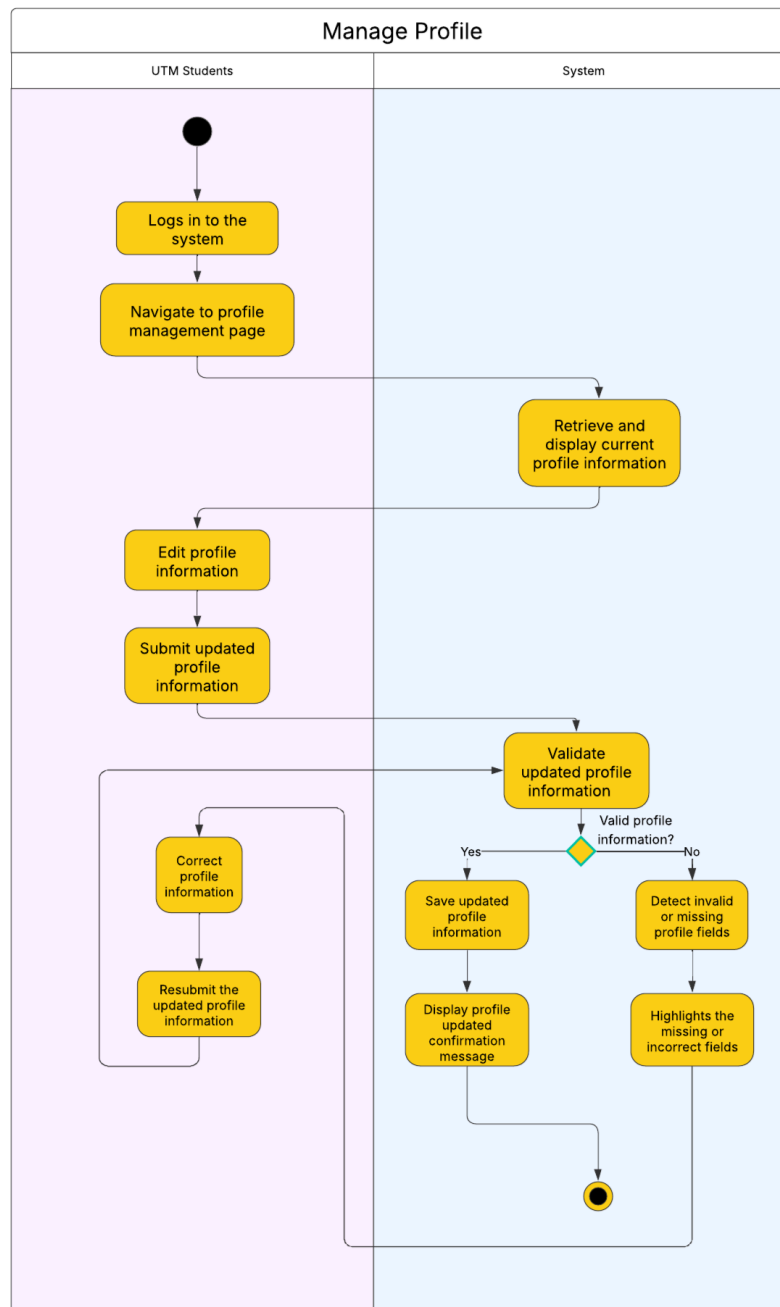
### 3. Manage Customer



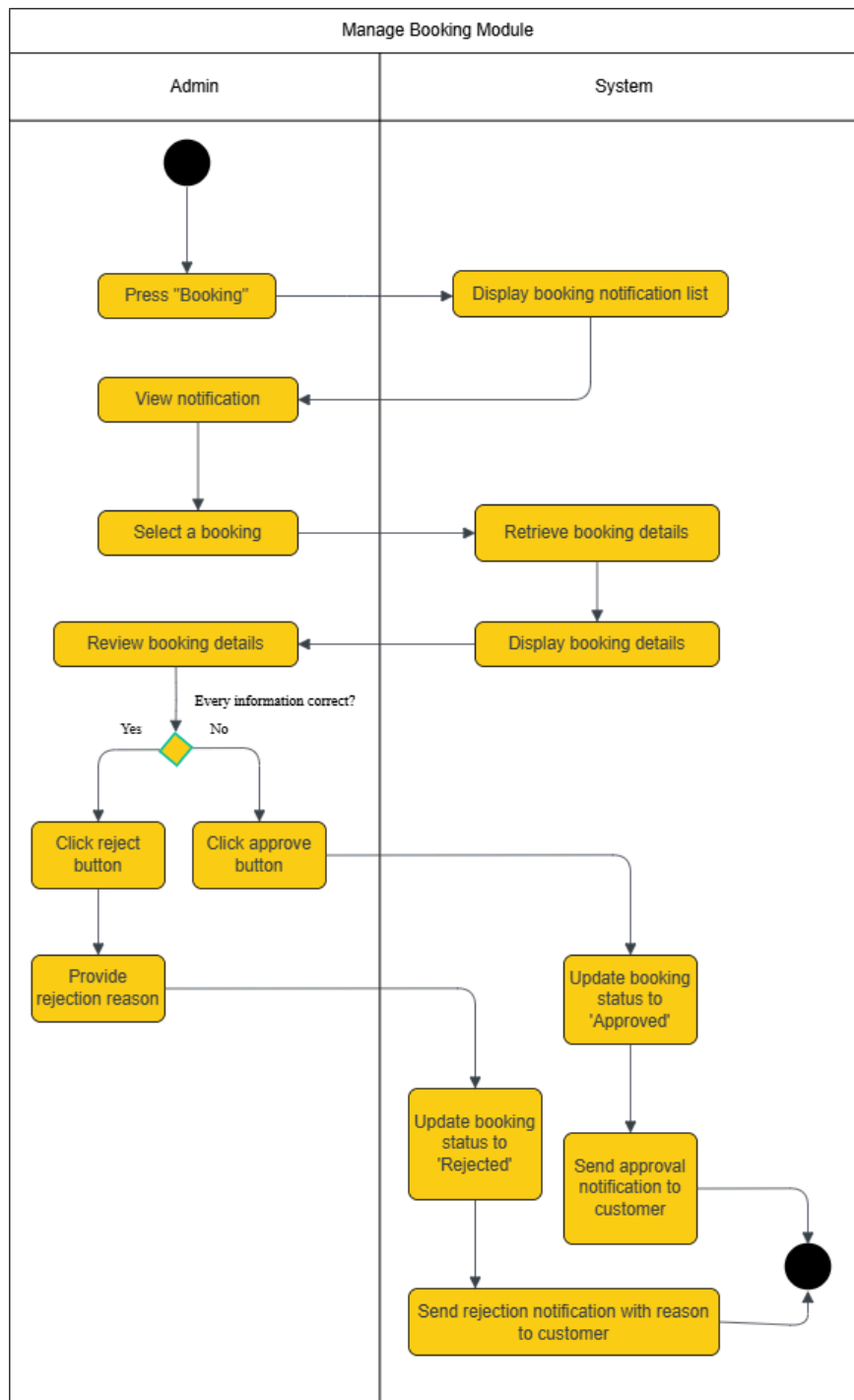
#### 4. Verify User Documents



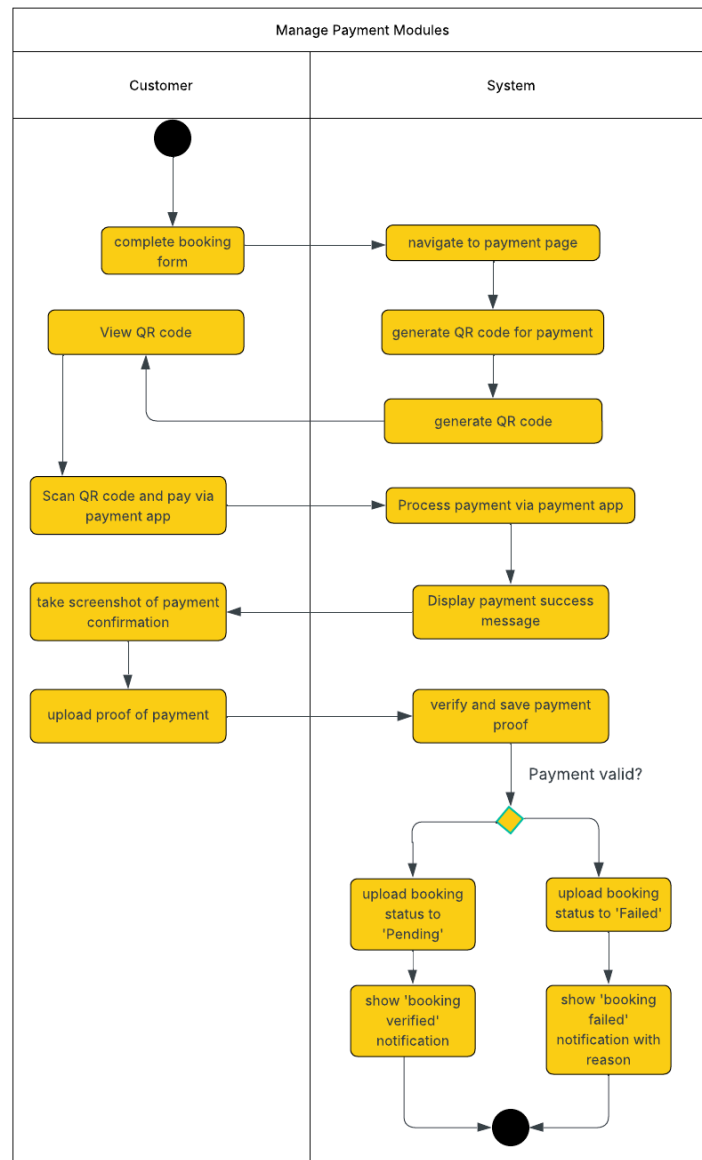
## 5. Manage Profile

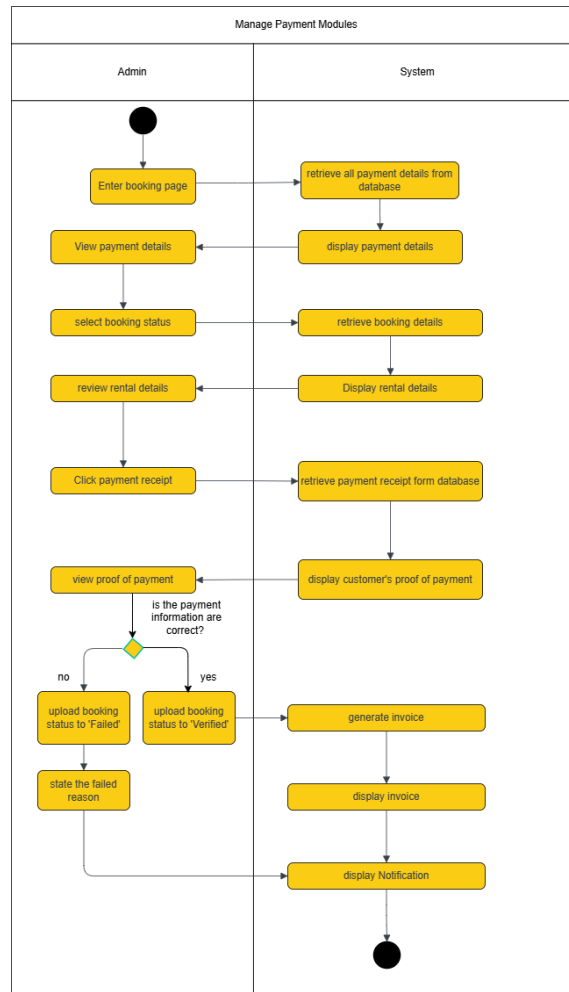


## 6. Manage Booking

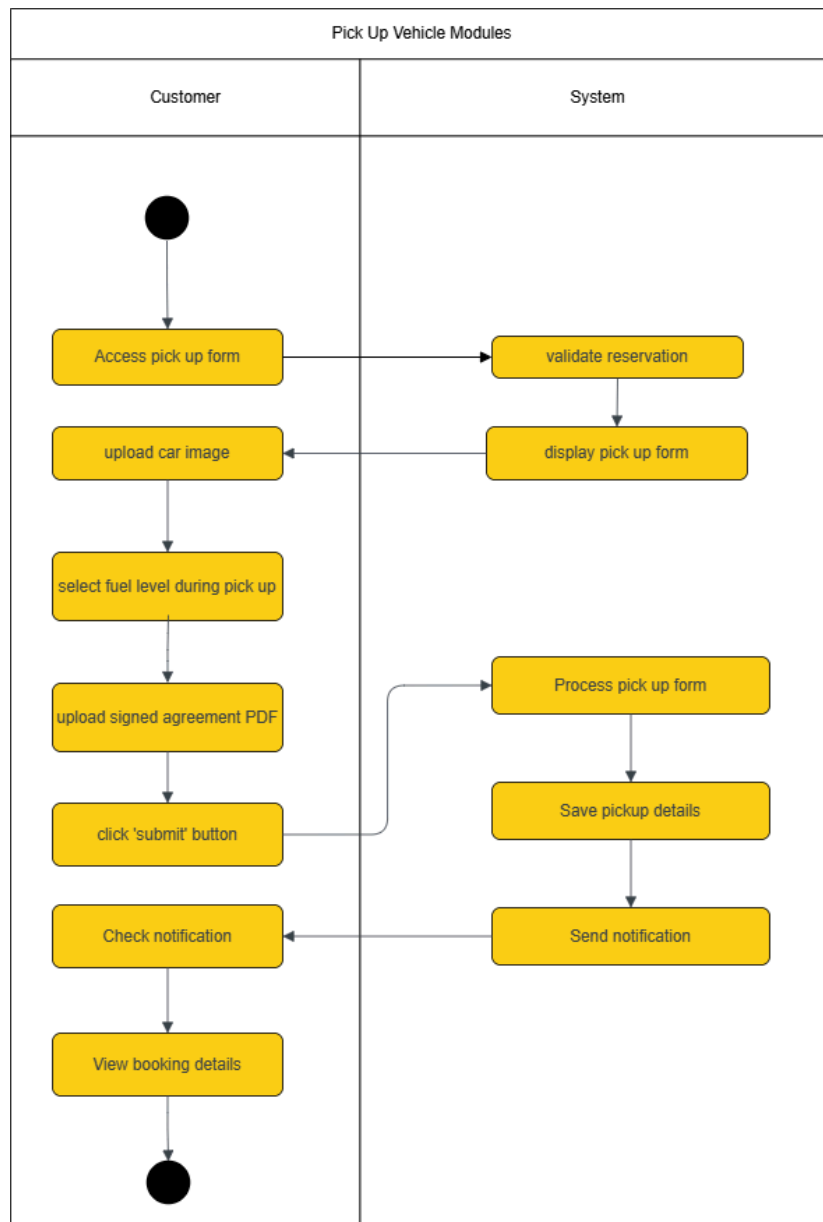


## 7. Manage Payment

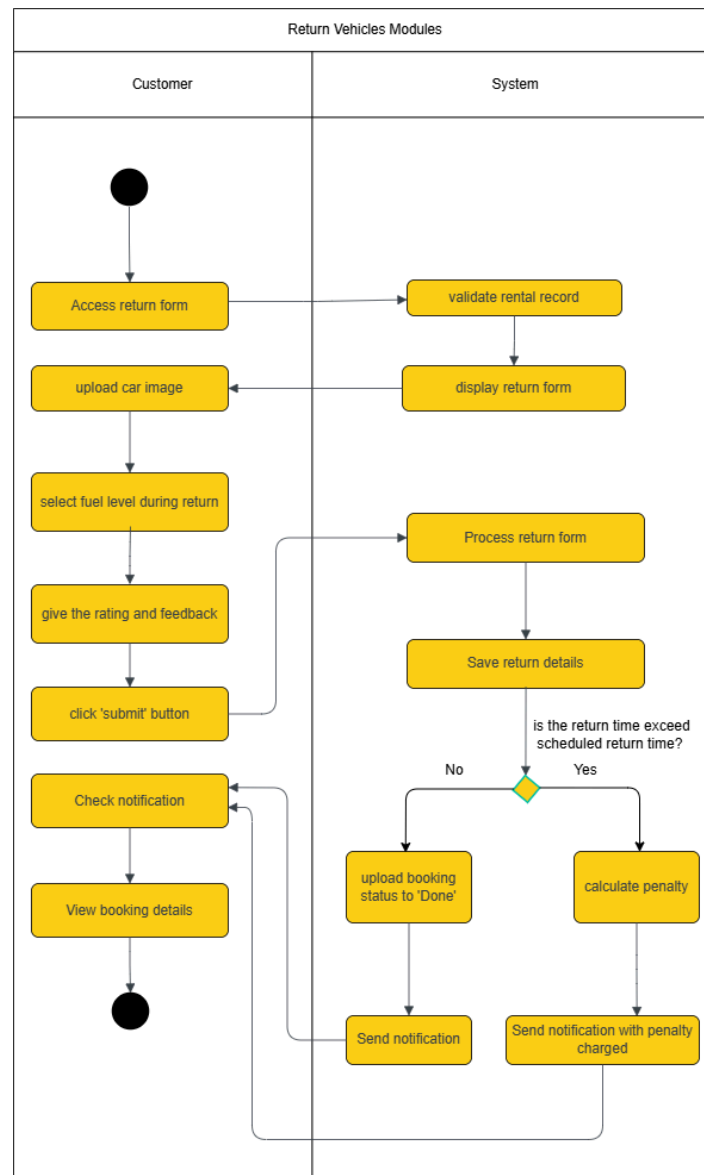




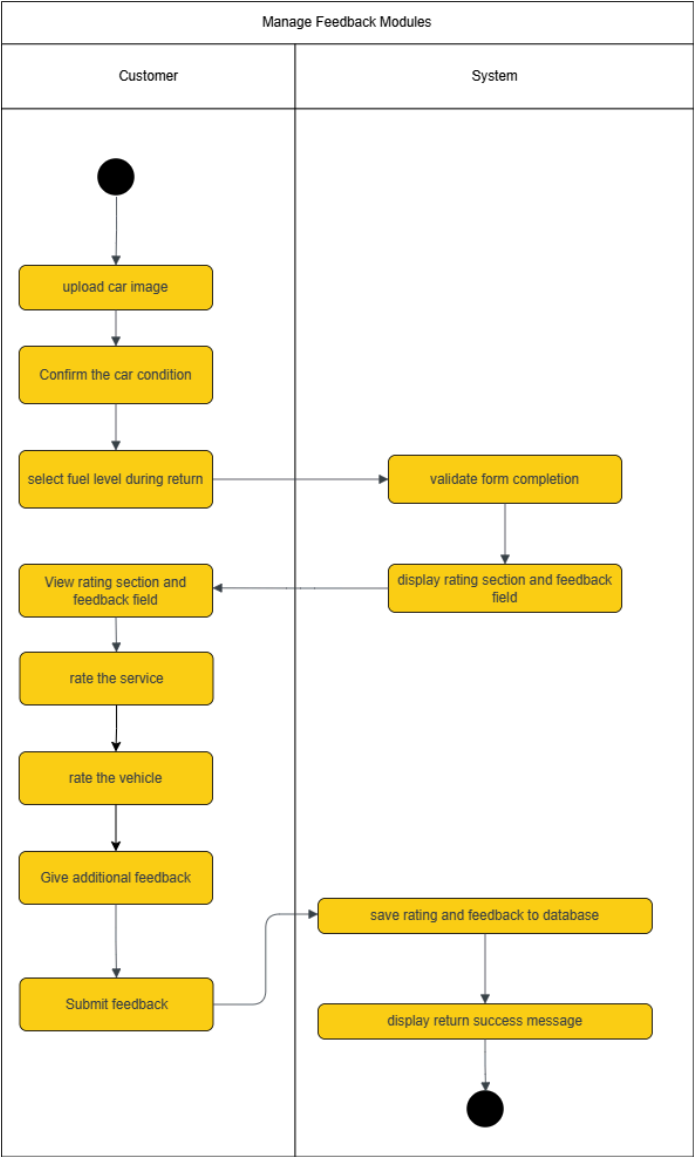
## 8. Pick Up Vehicle

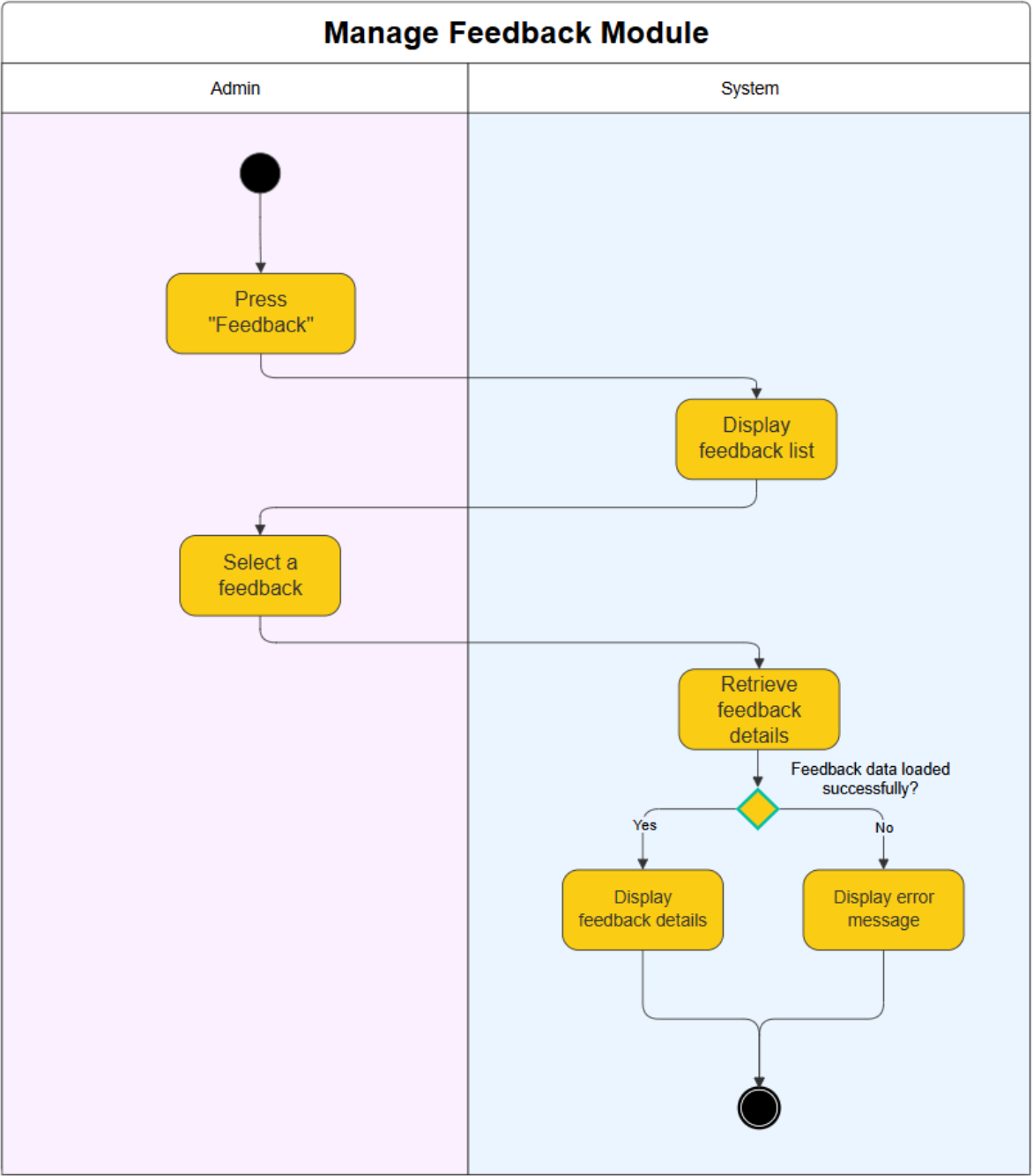


## 9. Return Vehicle

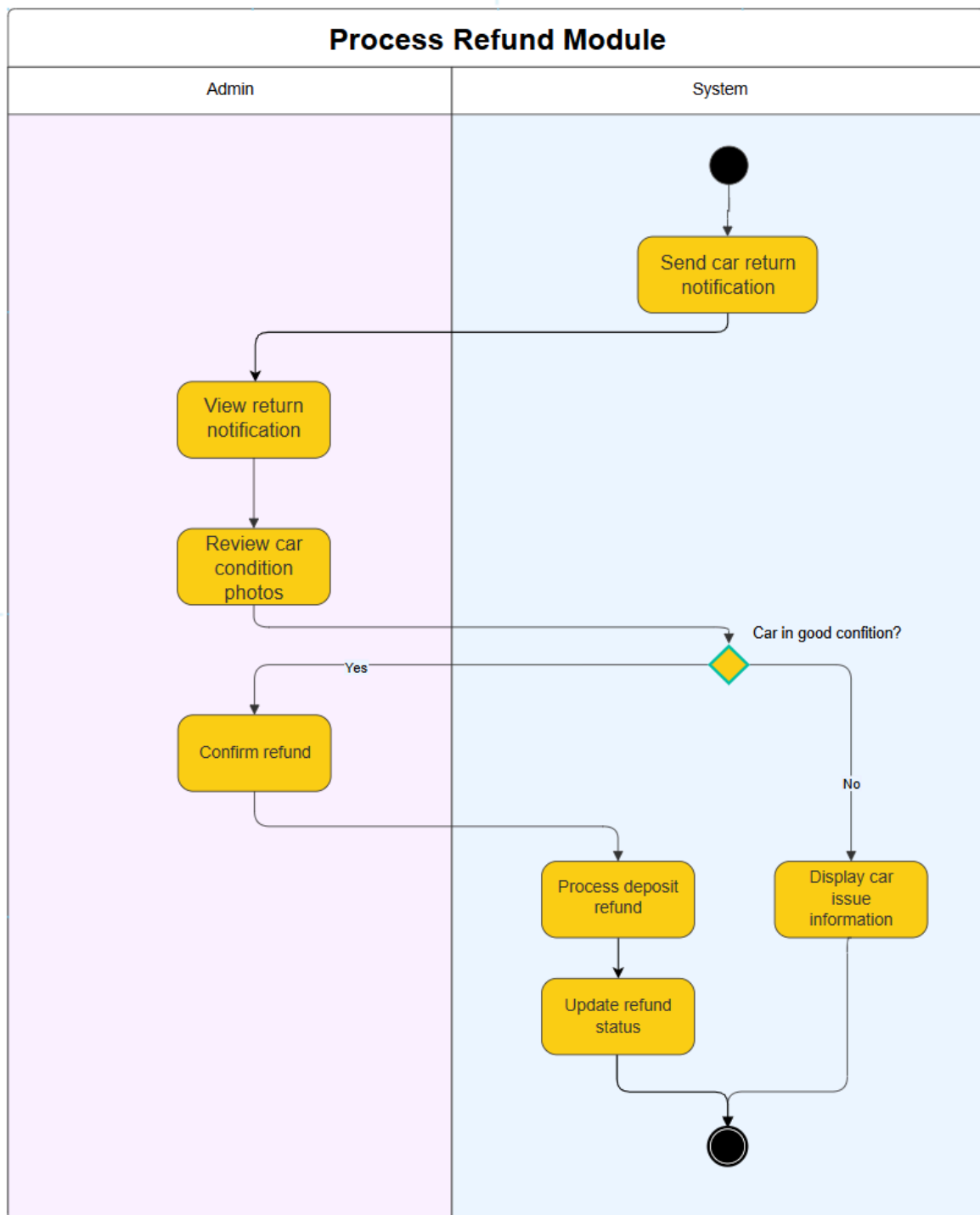


10. Manage Feedback

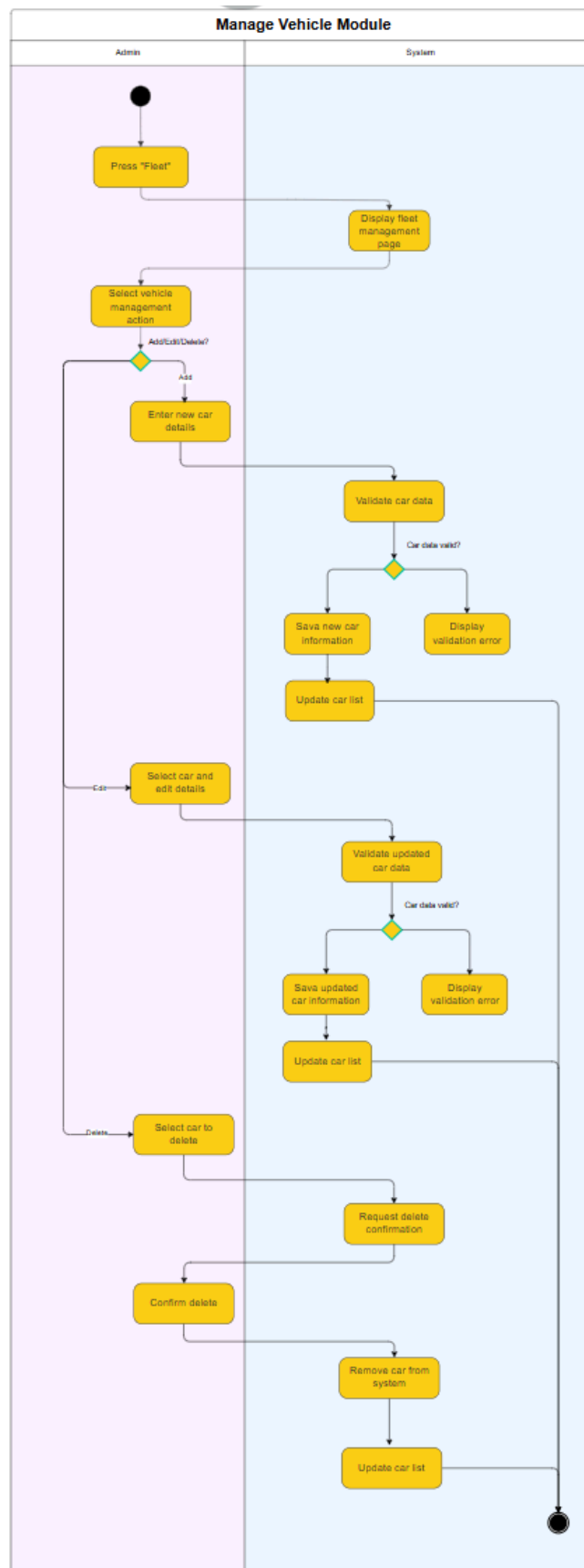




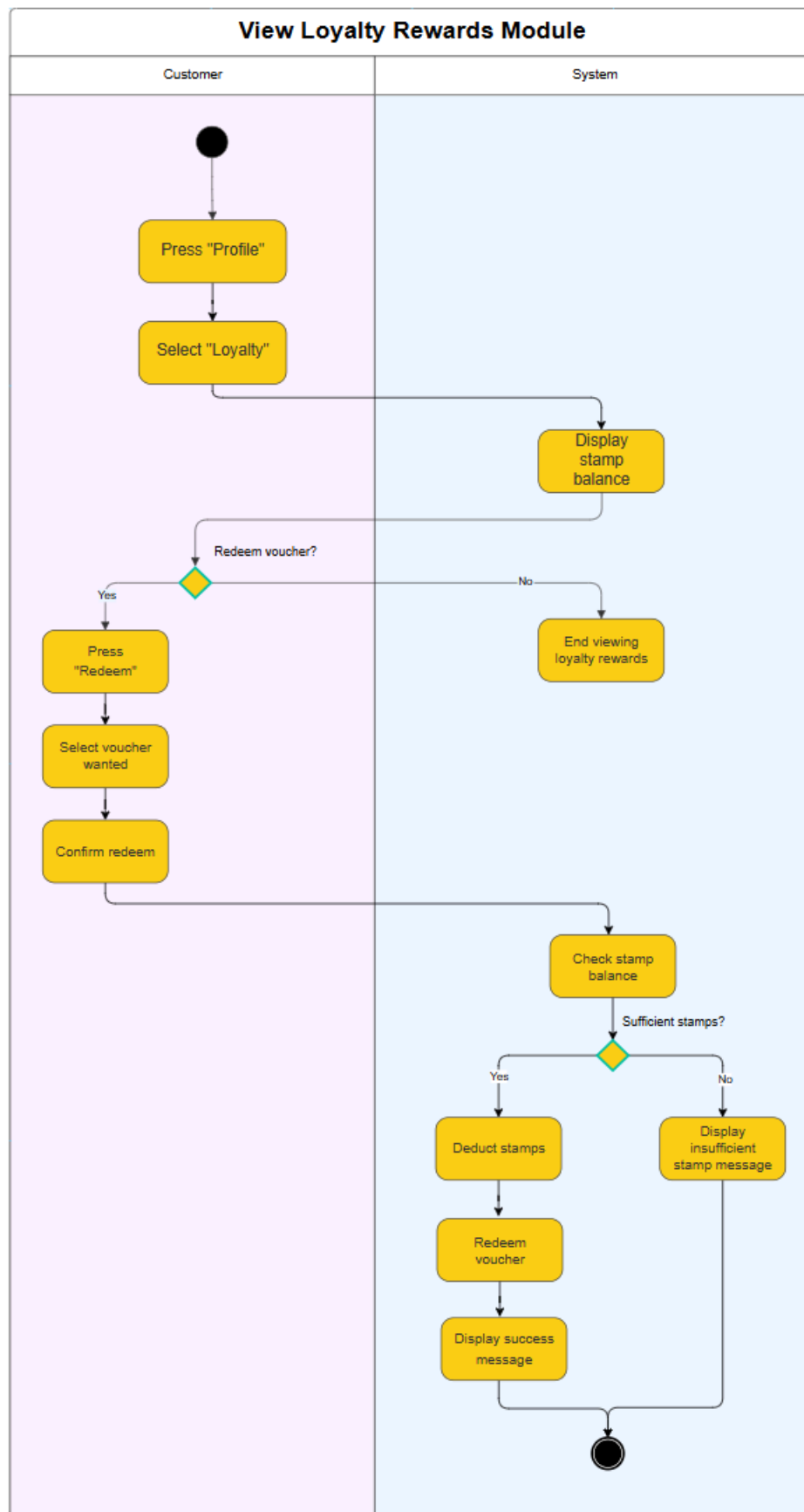
## 11. Process Refund



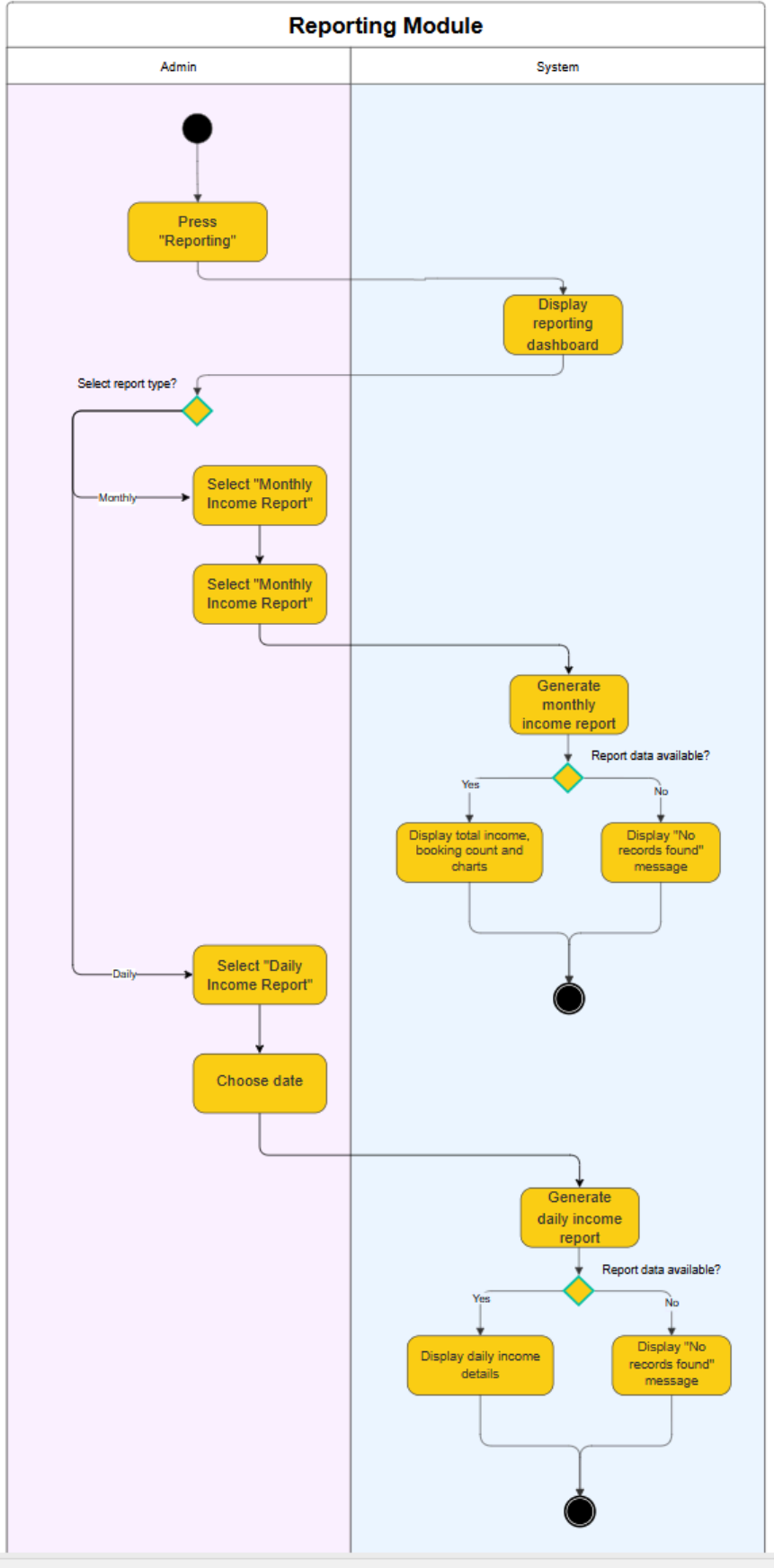
## 12. Manage Vehicle



### 13. View Loyalty Rewards



14. Reporting



### 7.3 Meeting Logs

#### MEETING MINUTES

|                        |                              |                                                                                                                                                         |                    |
|------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Date/Time              |                              | 14 January 2026 9:30pm                                                                                                                                  |                    |
| Location               |                              | Google Meet                                                                                                                                             |                    |
| Agenda                 |                              | 1. Review project requirements and scope for Hasta Travel Vehicle Rental Management System                                                              |                    |
|                        |                              | 2. Discuss and finalize system modules and their functionalities                                                                                        |                    |
|                        |                              | 3. Assign responsibilities for ERD development and documentation                                                                                        |                    |
|                        |                              | 4. Review Data Flow Diagrams (Context, Level 0, and Child Diagrams)                                                                                     |                    |
|                        |                              | 5. Establish timeline and milestones for deliverables                                                                                                   |                    |
| Meeting MC             |                              | Angela Ngu Xin Yi                                                                                                                                       |                    |
| Attendance             |                              |                                                                                                                                                         |                    |
| Name                   |                              | Time                                                                                                                                                    | Reason for Absence |
| Angela Ngu Xin Yi      |                              | 2130                                                                                                                                                    | -                  |
| Gavenesh A/L Batumalai |                              | 2130                                                                                                                                                    | -                  |
| Tan Xin Tian           |                              | 2130                                                                                                                                                    | -                  |
| Teoh Xin Yee           |                              | 2130                                                                                                                                                    | -                  |
| Toh Shee Thong         |                              | 2130                                                                                                                                                    | -                  |
| Minutes                |                              |                                                                                                                                                         |                    |
| No.                    | Item Discussed               | Ideas/Suggestions and Person Giving It                                                                                                                  | Person in Charge   |
| 1.                     | Project Scope and Objectives | Angela:<br>Emphasized the need to improve Hasta Travel's manual booking system to eliminate WhatsApp-based inefficiencies and provide real-time vehicle | Angela, Gavenesh   |

|    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |
|----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|    |                                    | <p>availability.</p> <p>Gavenesh:<br/>Suggested focusing on UTM-specific features like matric number verification and campus-based pickup locations.</p> <p>All members agreed on core features: automated booking, payment verification, loyalty rewards, and admin dashboard.</p>                                                                                                                                                       |                                  |
| 2. | System Modules Definition          | <p>Xin Tian:<br/>Proposed six main modules - User Management, Fleet Management, Booking Management, Payment Management, Loyalty &amp; Rewards, and Report &amp; Analytics.</p> <p>Xin Yee:<br/>Recommended separating pickup and return processes as distinct entities in the database for better tracking.</p> <p>Shee Thong:<br/>Suggested including vehicle maintenance tracking and renewal reminders for road tax and insurance.</p> | Xin Tian, Xin Yee and Shee Thong |
| 3. | ERD and Database Design Assignment | <p>Angela:<br/>Assigned ERD development tasks:</p> <ul style="list-style-type: none"> <li>- User Management Module: Xin Yee</li> <li>- Fleet Management Module: Xin Tian</li> <li>- Booking Management Module: Shee Thong</li> <li>- Payment Management Module: Gavenesh</li> <li>- Loyalty &amp; Rewards Module: Angela</li> <li>- Integration and Enhanced ERD: All members to review together</li> </ul>                               | Angela, Gavenesh                 |

|    |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |
|----|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
|    |                                | <p>Gavenesh:<br/>Suggested using specialization for User entity (Customer/Admin hierarchy).</p> <p>All agreed to maintain consistency in naming conventions and relationship cardinalities.</p>                                                                                                                                                                                                                                                                                            |                               |
| 4. | DFD Review and Validation      | <p>Xin Yee:<br/>Presented draft Context Diagram showing main external entities (Customer, Admin, Manager) and system boundary.</p> <p>Shee Thong:<br/>Reviewed Level 0 DFD covering 8 main processes from Login to Deposit Withdrawal.</p> <p>Xin Tian:<br/>Suggested adding data stores for each module to show data flow clearly.</p> <p>Group discussion:<br/>Refined child diagrams for complex processes like "Book the Car" and "Verify Booking" to show detailed sub-processes.</p> | Xin Yee, Shee Thong, Xin Tian |
| 5. | Business Rules and Constraints | <p>All members brainstormed and documented business rules</p> <p>Xin Yee:<br/>User verification requirements - customers must upload IC, matric card, and driving license before booking.</p> <p>Xin Tian:<br/>Vehicle availability logic - maintenance status overrides booking availability.</p> <p>Shee Thong:<br/>Booking constraints -</p>                                                                                                                                            | All members                   |

|  |  |                                                                                                                                                                                                                                                                                                                  |  |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | <p>24-hour cancellation policy,<br/>30-minute grace period for late<br/>returns.</p> <p>Gavenesh:<br/>Payment rules -<br/>QR payment only, manual admin<br/>verification required.</p> <p>Angela:<br/>Loyalty program -<br/>1 stamp per rental <math>\geq 9</math> hours, stamp<br/>redemption for vouchers.</p> |  |
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